

Enantioselective Addition of Organometallic Reagents to Carbonyl Compounds: Chirality Transfer, Multiplication, and Amplification**

New Synthetic
Methods (83)

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Nucleophilic addition of organometallic reagents to carbonyl substrates constitutes one of the most fundamental operations in organic synthesis. Modification of the organometallic compounds by chiral, nonracemic auxiliaries offers a general opportunity to create optically active alcohols, and the catalytic version in particular provides maximum synthetic efficiency. The use of organozinc chemistry, unlike conventional organolithium or -magnesium chemistry, has realized an ideal catalytic enantioselective alkylation of aldehydes leading to a diverse array of secondary alcohols of high optical purity. A combination of dialkylzinc compounds and certain sterically constrained β -dialkylamino alcohols, such as (–)-3-*exo*-dimethylaminoisoborneol [(–)-DAIB], as chiral inducers affords the best result (up to 99% *ee*). The alkyl transfer reaction occurs via a dinuclear Zn complex containing a chiral amino alkoxide, an aldehyde ligand, and three alkyl groups. The chiral multiplication method exhibits enormous chiral amplification: a high level of enantioselection (up to 98%) is attainable by use of DAIB in 14% *ee*. This unusual nonlinear effect is a result of a marked difference in chemical properties of the diastereomeric (homochiral and heterochiral) dinuclear complexes formed from the dialkylzinc and the DAIB auxiliary. This phenomenon may be the beginning of a new generation of enantioselective organic reactions.

1. Introduction

Chirality is a major phenomenon in Nature, and molecular asymmetry in particular is playing a crucial role in science and technology. A variety of significant biological functions emerge through molecular recognition, which requires strict matching of chirality. Nonracemic molecules will also contribute to the creation of advanced materials possessing unique physical properties. Accordingly enantioselective synthesis of chiral organic compounds is an important task allotted to synthetic chemists. Asymmetric catalysis using chiral metal complexes, among others, provides a general, powerful tool in this context.^[1]

The carbon–carbon bond forming reaction, together with oxidation, reduction, and functional group transformation, is one of the most fundamental operations for constructing organic molecules. Addition of organometallic reagents to carbonyl compounds is among the most common reactions for this purpose, complementary to reduction of aldehydes or ketones, and provides a reliable way to form alcoholic products. Organometallic chemistry dates back to the discovery of organozinc compounds by *Frankland* in 1849.^[2] Although this pioneering achievement was deeply appreciated, the significance of organozinc chemistry remained somewhat underestimated, at least in terms of synthetic utility. The Reformatsky reaction^[3] and Simmons–Smith reaction^[4] utilize Zn metal but, in simple alkylation to carbonyl compounds, organozinc compounds were replaced by Grignard reagents (discovered 50 years later)^[5] and organolithium compounds (developed by *Wittig* and *Gilman*)^[6] among

others), owing perhaps to their higher generality, higher reactivity, and easier manipulation. Recently, the chemistry of organoboron, -aluminum, -titanium, -lanthanide compounds, etc. has been further developed with the hope of obtaining high selectivities. The enantioselective version of the reaction between organometallic reagents and carbonyl compounds which leads to optically active alcohols is desirable because of its general synthetic importance.

A nonracemic, chiral environment can be introduced into organometallic compounds in two ways (Fig. 1). One is the

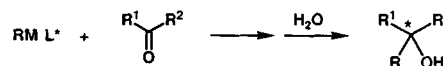


Fig. 1. Enantioselective addition of organometallic reagents to carbonyl compounds. L* = neutral or anionic chiral ligand

coordination of aprotic chiral solvents or complexing agents to the metallic center, and the other is the modification of the organometallic reagent by protic chiral ancillaries, such as alcohols or amines, to give organometallic alkoxides or amides, respectively. The molecular structures of various main-group organometallic compounds having neutral and anionic ligands have been studied by X-ray crystallographic analysis and spectroscopic methods, revealing that the steric and electronic properties of the ligands significantly affect the structures, including configuration, metal coordination number, and bond polarity.^[7–9] Many organometallic compounds form labile molecular associates, so the aggregation state responsible for the reaction may be different from that of the ground-state structure. The organometallic compounds are amphiphilic in nature: the nucleophilicity of the carbon moiety and Lewis acidity of the metallic center cooperate in the alkyl transfer reaction. Thus chiral ligands per-

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Ar represents aryl substituents in this article.

turb the structure and stability of the transition state in many ways.

In the last two decades, a number of highly enantioselective alkylation reactions of carbonyl compounds have been developed by proper combination of organometallic reagents and well-designed chiral ancillaries. So far most selective reactions have used a chiral auxiliary in a stoichiometric or an excess amount, but it has been our hope to realize a catalytic asymmetric induction. In this respect, a number of unexpected, nonclassical aspects have been observed with the classical organozinc chemistry. This review describes the historical background and our personal account of this research.^[10–13] The subject is limited to simple alkylation or arylation reactions. The chemistry using structurally more elaborate carbon nucleophiles, such as metal enolates or allylic metal compounds, is not included.

2. Stoichiometric Asymmetric Alkylation

2.1. Reaction with Organometallic Reagents Modified by Aprotic Chiral Ligands

The curtain on enantioselective alkylation of aldehydes was raised in 1940 by *Betti* and *Lucchi* who claimed the first successful use of Grignard reagents and a chiral tertiary amine.^[14] The Bologna chemists reported that the reaction of benzaldehyde and methylmagnesium iodide in the presence of *N,N*-dimethylbornylamine gave, in 73% yield, 1-

phenylethanol showing an optical rotation of 0.30°. However, two years later *Tarbell* and *Paulson*, who were unable to reproduce this result, concluded that the optically active alcohol had not been produced although a neutral optically active impurity was formed by the reaction.^[15] The original concept remained valid, however: certain chiral complexing agents are able to effect enantioselective addition of organometallic compounds to aldehydes.

Cohen and *Wright* reported in 1953 that the reaction of ethyl 2-oxo-2-phenylacetate with 1.6 equivalents of ethylmagnesium chloride and (2*R*,3*R*)-dimethoxybutane in benzene yielded ethyl 2-hydroxy-2-phenylbutanoate in 5% *ee* and in 37% yield.^[16] Multidentate ethers such as dimethoxybutane, hexa(*O*-methyl)mannitol, or penta(*O*-methyl)arabitol are more effective than unidentate chiral ethers.^[17] Table 1 outlines the development of this type of chemistry.

In the late 1960's we became interested in the use of (–)-sparteine **1**, a naturally occurring tertiary diamine, and noticed a 22% *ee* in the reaction of benzaldehyde and ethylmagnesium bromide.^[18] Organolithium reagents were less sensitive than Grignard reagents to the presence of this chiral ligand. *Seebach* studied systematically the effects of multidentate ligands^[19] and attained 52% *ee* in the reaction of benzaldehyde and *n*-butyllithium aided by an amino ether **2**.^[19d] Ultimately, *Mazaleylat* and *Cram* accomplished 95% *ee* in the same alkylation using the chiral diamine ligand **3** having three *C*₂ axes.^[20] A strong solvent effect was observed and use of diethyl ether at –120°C gave the best *ee*

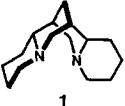
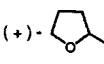
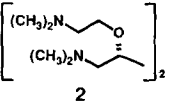
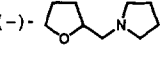
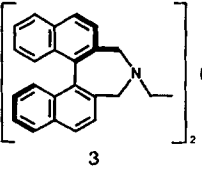
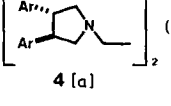


Ryoji Noyori, born in Kobe, Japan in 1938, completed his undergraduate study in 1961 and the Master's degree in 1963 at Kyoto University, and immediately became an instructor in Professor H. Nozaki's laboratory at the same university. While at Kyoto he earned his Ph.D. degree in 1967. Appointed as an associate professor at the Department of Chemistry of Nagoya University in 1968, he was promoted to his current position in 1972. Noyori did his postdoctoral work with Professor E. J. Corey at Harvard University in 1969–1970. His research interests have been mainly in the exploitation of new synthetic methods, particularly on the basis of organometallic chemistry and their application. Noyori discovered a variety of asymmetric syntheses including enantioselective catalyses using BINAP (p,p'-(1,1'-binaphthyl-2,2'-diyl)bis(diarylphosphane)) chemistry, which have enormous utility. His organometallic methodologies have opened truly efficient routes to terpenes, vitamins, β-lactam antibiotics, alkaloids, prostaglandins, carbohydrates, nucleic acids, etc. Noyori's contribution has been recognized by a number of awards and honors, which include the Chemical Society of Japan Award (1985), the Centenary Medal and Lectureship from the Royal Society of Chemistry (1989), the Fluka Prize (1989), the Naito Foundation Research Prize (1989), the Toray Science & Technology Prize (1990).



Masato Kitamura was born in Nagoya in 1956 and studied agricultural chemistry at Nagoya University where he completed his doctoral work on total synthesis of maytansine under the direction of Professors T. Goto and M. Isobe. In 1983 he was a postdoctoral fellow with Professor G. Stork (Columbia University) and then joined the Noyori group as an assistant professor. He is a recipient of the Chemical Society of Japan Award for Young Chemists (1989).

Table 1. Chiral ligand aided enantioselective addition of organometallic reagents to carbonyl compounds.

Chiral ligand (equiv.)	Reagent (equiv.)	Substrate (equiv.)	Optical yield, configuration	Ref.
 (1)	C_2H_5MgCl (1)	$C_6H_5COCO_2C_2H_5$ (0.64)	5, –	[16]
 (1)	C_2H_5MgBr (1)	C_6H_5CHO (0.8)	22, (R)	[18]
(+)-  (1)	C_6H_5MgBr (0.22)	<i>t</i> -C ₄ H ₉ CHO (0.33)	11, –	[22]
 (1)	<i>n</i> -C ₄ H ₉ Li (0.55)	C_6H_5CHO (0.5)	52, (S)	[19d]
(-)-  (1)	CH_3Li (0.83)	C_6H_5CHO (0.75)	19, –	[23]
 (1)	<i>n</i> -C ₄ H ₉ Li (0.88)	C_6H_5CHO (0.24)	95, (R)	[20]
 (1)	1-C ₁₀ H ₇ MgBr (0.83)	C_6H_5CHO (0.33)	75, (S)	[21]

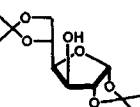
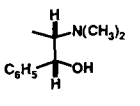
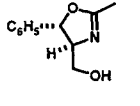
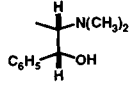
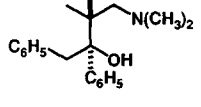
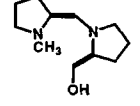
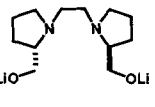
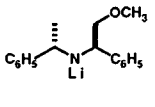
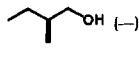
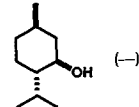
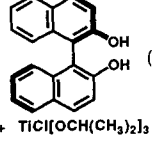
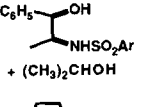
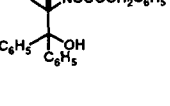
[a] Ar = 3,5-xylyl.

value. Similar diamine ligands, such as **4**, containing two *trans*-2,4-diarylpyrrolidine units were effective for asymmetric addition of arylmagnesium bromides to aldehydes, providing the corresponding alcohols in up to 75% *ee*.^[21]

2.2. Reaction with Organometallic Reagents Modified by Protic Chiral Auxiliaries

This strategy, though now widely utilized, was found rather recently. Since the importance of molecular aggregates in organometallic reactions was not correctly recognized, most synthetic chemists considered that the reaction of organolithiums or Grignard reagents of type RMgX and protic compounds formed, by eliminating an alkane, metallic compounds with no alkylation ability. In 1969, however, *Inch* et al. demonstrated that, in the presence of an α -D-glucofuranose derivative **5** containing a free hydroxyl group (Table 2), methylmagnesium bromide reacts with cyclohexyl phenyl ketone (alcohol:CH₃MgBr:ketone = 1:1.75:0.5) to afford the corresponding alcohol in 70% *ee* and in 95% chemical yield.^[24] This reaction probably involves a multi-

Table 2. Enantioselective reactions of chirally modified organometallic reagents and carbonyl compounds.

Chiral auxiliary (equiv.)	Reagent (equiv.)	Substrate (equiv.)	<i>ee</i> [%], configuration	Ref.
 (1)	CH_3MgBr (1.75)	$C_6H_5COCO_2H_{11}$ (0.5)	70, (R)	[24]
 (1)	$(C_2H_5)_2Mg$ (1)	<i>t</i> -C ₄ H ₉ CHO (0.75)	20, (R)	[25]
 (1)	CH_3MgBr (1.75)	$C_6H_5COCO_2H_5$ (0.5)	25, (S)	[29]
 (–)	$(CH_3)_2CuLi$ (–)	C_6H_5CHO (–)	16.5, (R)	[26]
 (1)	$(n-C_4H_9)_4AlLi$ (1)	$C_6H_5COCO_2CH_3$ (0.95)	43, –	[28]
 (1)	<i>n</i> -C ₄ H ₉ Li (1.68)	C_6H_5CHO (0.25)	95, (S)	[30]
 (1)	<i>n</i> -C ₄ H ₉ Li (0.67)	C_6H_5CHO (0.33)	36, (R)	[35]
 (1)	<i>n</i> -C ₄ H ₉ Li (0.68)	C_6H_5CHO (0.25)	90, (S)	[40]
 (–)	$CH_3Li/TiCl_4$ (–)	C_6H_5CHO (–)	8, (S)	[32a]
 (–)	$(CH_3)_2Ti(O-i-C_3H_7)_2$ (–)	C_6H_5CHO (–)	13.5, (S)	[32b]
 (R)- 11 (1)	C_6H_5MgBr (1)	1-C ₁₀ H ₇ CHO (0.77)	> 98, –	[33]
 (1)	$(CH_3)_4Ti$ (1)	<i>o</i> -NO ₂ C ₆ H ₄ CHO (0.91)	90, (R)	[34]
 (1)	$(CH_3)_2Ti(O-i-C_3H_7)_2$ (1)	C_6H_5CHO (0.33)	54.1, (R)	[36]

nuclear magnesium compound containing methyl group(s) and a sugar alkoxide.

Later, the influence of chiral amino alcohols such as *N*-methylephedrine **6**^[25–27] and Darvon alcohol **8**^[28] or chiral dihydrooxazoles such as **7**^[29] was examined in alkylations using Grignard reagents, organocuprates, organoaluminates, etc. *Mukaiyama* attained 95% *ee* in the reaction of benzaldehyde and *n*-butyllithium in the presence of a proline-derived diamine, **9**, in a 1:1 mixture of dimethoxy-methane and dimethyl ether at –123 °C.^[30] The reaction using lithium trimethylsilylacetylide also proceeded in 92% optical yield and was used for the synthesis of corticoids.^[31]

Chirally modified organotitanium compounds are also useful.^[32] *Seebach* et al., after extensive experimentation, found that aromatic aldehydes are arylated with organotitanium compounds generated in situ from enantiomerically pure 2,2'-dihydroxy-1,1'-binaphthyl **11** (binaphthol), chlorotriisopropoxytitanium and aryl Grignard reagents in THF to give the diaryl alcohols in greater than 98% *ee* in some cases.^[33] *Reetz* et al. noted that *N*-sulfonylated derivatives of norephedrine, such as **12**, are good chiral ligands for modification of methyltitanium reagents, which effect alkylation of aromatic aldehydes to form (*R*)-alcohols in ca. 90% *ee*.^[34] ¹H and ¹³C NMR spectra of the titanium reagent showed a multitude of peaks not in line with a single monomeric or dimeric species. The results of several representative examples of reactions with reagents modified by chiral ligands are given in Table 2.

2.3. Reaction with Lithium/Magnesium Binary Reagents Modified by Binaphthol

A high level of enantioselection, based on ligand-directed chirality transfer, is obtained by careful combination of chiral auxiliaries, carbonyl compounds, and organic groups (alkyl, aryl, etc.), as exemplified in Section 2.2. A general method did not exist. The chemical and optical yields of reactions using organolithium or -magnesium compounds often vary depending on the reaction conditions, the ratio chiral ligand: alkylmetal: substrate, the way the chiral reagents were prepared, etc.

The use of excess amounts of chiral sources (usually 4 equivalents or more) over carbonyl substrates is imperative in many cases in order to achieve a high level of enantioselective addition, although the titanium reagents were usually utilized with only 1 to 1.3 equivalents of chiral auxiliaries.^[33] This can be a result of the competition between achiral and chiral alkylating agents present in the reaction system, leading to products with varying selectivities and at different rates. Therefore, the creation of a single highly reactive species possessing excellent chiral recognition ability is crucial in achieving a stereoselective reaction. A readily available Li/Mg binary organometallic agent **13** (Fig. 2, empirical formula) is advantageous for this purpose: coordinatively saturated tetrahedral lithium and magnesium can avoid the complication of reagents by polymerization or aggregation, and the C₂ chiral bidentate modifier can minimize the number of the reactive species.^[33, 37] The two kinds of metals may play specific roles as alkyl donor and carbonyl activator, which results in high reactivity.

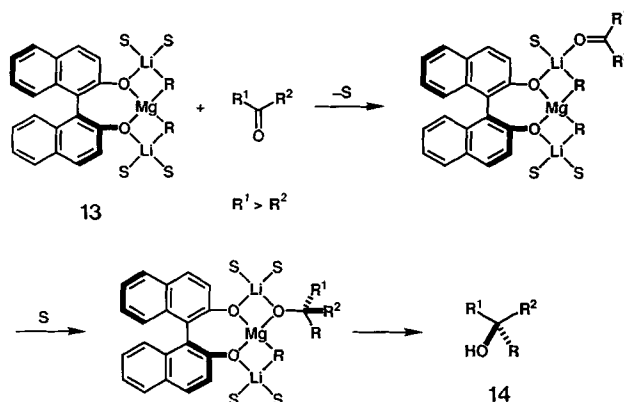


Fig. 2. Enantioselective addition of binaphthol modified Li/Mg binary organometallic agents to carbonyl compounds. S = solvent.

Thus the ethylating agent, (*S*)-**13** (*R* = ethyl), possessing two homotopic ethyl groups, was prepared by treatment of enantiomerically pure (*S*)-binaphthol (*S*)-**11** dissolved in THF with 2 equivalents of *n*-butyllithium, and then with 1 equivalent of diethylmagnesium at –78 °C. And indeed, the golden-yellow homogeneous reagent (0.074 M) reacted with 1 equivalent of benzaldehyde at –100 °C for 1 h in a 1:1 mixture of THF and dimethoxyethane (DME) to form (*S*)-1-phenyl-1-propanol (**14**, *R* = ethyl, *R*¹ = phenyl, *R*² = H) in 92% *ee* and 93% yield. As exemplified in Table 3, a variety of aromatic or aliphatic aldehydes were alkylated in greater than 80% optical yield with the alkyl or phenyl reagents using a 1:1 stoichiometry (not an excess).^[11]

Table 3. Enantioselective reactions of the Li/Mg reagent, (*S*)-**13**, with carbonyl compounds R¹R²C=O [a].

Organo-metallic R in (<i>S</i>)- 13	R ¹	R ²	Yield [%]	<i>ee</i> [%]	Configuration
C ₂ H ₅	C ₆ H ₅	H	93	92	<i>S</i>
C ₂ H ₅	<i>p</i> -CH ₃ OC ₆ H ₄	H	98	83	<i>S</i>
C ₂ H ₅	C ₆ H ₅ CH ₂ CH ₂	H	40	85	<i>S</i>
C ₂ H ₅	<i>n</i> -C ₆ H ₁₃	H	86	85	<i>S</i>
C ₂ H ₅	<i>c</i> -C ₆ H ₁₁	H	68	84	[b]
<i>n</i> -C ₄ H ₉	C ₆ H ₅	H	98	88	[b]
C ₆ H ₅	1-C ₁₀ H ₇	H	89	82	[b]
C ₂ H ₅	C ₆ H ₅	CH ₃	57	45	[b]

[a] The reaction was carried out at –100 °C in a 1:1 THF–DME mixture.
[b] Not determined.

A 0.074 M [*D*₈]THF solution of di-*O*-lithiobinaphthol, prepared by mixing (*S*)-**11** and *n*-butyllithium in a 1:2 mole ratio followed by recrystallization from THF–hexane, exhibited a broad ⁷Li NMR signal at δ = –0.25 (LiCl in [*D*₈]THF as external reference, half-width *v*_{1/2} = 14 Hz at 20 °C). The aromatic protons also gave broad NMR signals. However, addition of 1 equivalent of diethylmagnesium to this solution sharpened the signals to a great extent (Fig. 3).

The ¹H NMR spectra of the ethylating agent **13** (*R* = ethyl) revealed a very sharp, single triplet and quartet due to the methyl and methylene protons at δ = 1.32 and –0.92, respectively, at temperatures ranging from 20 to –50 °C.^[38] The ⁷Li NMR spectrum, measured at 20 °C,

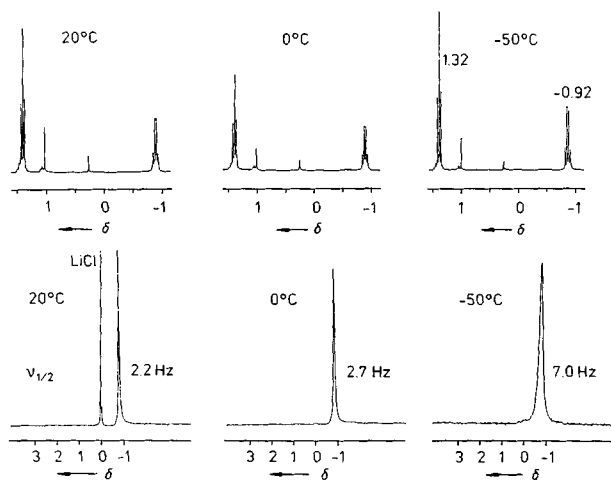
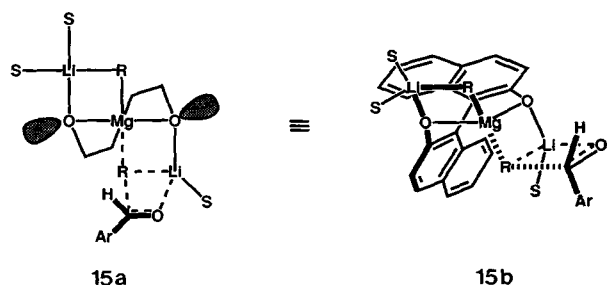


Fig. 3. ^1H (top, THF as internal standard) and ^7Li NMR spectra (bottom, LiCl in $[\text{D}_8]\text{THF}$ as external standard) of the Li/Mg ethylating agent **13** ($\text{R} = \text{C}_2\text{H}_5$, $\text{S} = [\text{D}_8]\text{THF}$, 0.074 M $[\text{D}_8]\text{THF}$ solution) at 20, 0, and -50°C .

showed a single signal ($\nu_{1/2} = 2.2$ Hz) at $\delta = -0.96$, and this signal was broadened at -50°C ($\delta = -0.98$, $\nu_{1/2} = 7.0$ Hz). Obviously some equilibration such as exchange of the coordinated THF molecules exists, but such a dynamic effect does not affect the magnetic environment of the ethyl protons to any great extent. Upon addition of benzaldehyde the spectrum became complicated, giving several new broad ^7Li signals at lower field. Notably, however, the original $\delta = -0.98$ signal due to **13** remained, though with decreased intensity, during the alkylation reaction.

The degree of enantioselection in alkylations with **13** was little affected by the extent of conversion of the starting materials. The ethylation of benzaldehyde using 4 equivalents of (*S*)-**13** ($\text{R} = \text{ethyl}$) afforded the same enantioselectivity ($S/R = 96:4$) as 1 equivalent, while the reaction with half an equivalent of the reagent resulted in decrease in the selectivity, $S/R = 91:9$ (90% yield). The resulting alkylmetal alkoxide products may cause adverse enantioselective alkylation.^[39]

The enantioselectivity of the ethylation of benzaldehyde varies from 66% to 92% *ee* on changing the reaction temperature from 20 to -100°C . Plotting of $\ln(S/R)$ as a function of $1/T$ shows direct proportionality, and from the line slope the difference in activation energy of the alkylation leading to the *S* and *R* alcohols is calculated to be 1.3 kcal/mol. This linearity suggests the participation of a single chiral reagent in the enantioselective alkylation over this temperature range. A further increase of enantioselectivity is expected by decreasing the reaction temperature, but the



freezing of the reaction media prohibits such experiments. A possible transition state giving the *S* alkoxide product is illustrated by the structure **15** (naphthalene rings are omitted in **15a**).

The reaction with (*S*)-**13** provides one of the most effective ways to alkylate carbonyl compounds in an enantioselective manner, but it is still based on stoichiometric chirality transfer; the preparation of optically active alcohols suffers from the necessity of using an equimolar amount of binaphthol.

3. Catalytic Enantioselective Alkylation

3.1. The Concept

Reactivity of organometallic compounds is profoundly affected by neutral or anionic heteroatom auxiliaries. In certain cases, addition of a chiral auxiliary could enhance the reactivity and at the same time control the stereochemical outcome in an absolute sense. Catalytic asymmetric alkylation is the obvious ideal and Figure 4 presents a possible pathway to this ultimate goal, where addition of dialkylmetal, R_2M , to a prochiral carbonyl substrate is achieved by a catalytic amount of protic chiral auxiliary, HX^* . Actually the metallic compounds are not simple monomers as formulated but usually exist as aggregates or in forms associated with other molecules.^[9]

Fig. 4. Enantioselective alkylation catalyzed by protic auxiliary HX^* . M = metallic species. X^* = chiral heteroatom ligand.

In order to obtain a high degree of enantioselection in this scheme, the chiral anionic ligand X^* must possess a suitable three-dimensional structure which clearly differentiates between the diastereomeric transition states of the alkyl delivery step, $16 \rightarrow 17$. In addition, several kinetic conditions are to be satisfied to achieve catalytic asymmetric induction. First, the rate of the alkylation by the chirally modified organometallic **16** should substantially exceed that of the reaction of the original achiral reagent R_2M . Second, ligand X^* must be readily detached from the initially formed metal alkoxide **17** by the action of the alkyl donor or carbonyl substrate to establish the catalytic cycle. These are key issues for obtaining a high turnover efficiency and an excellent chiral bias. These considerations also apply to the reaction using aprotic modifiers.

Although a wide array of well-shaped chiral auxiliaries or ligands are now accessible, the kinetic requirements described above are not easily secured by use of conventional organolithium or -magnesium compounds. *Cram* has found

that reaction of *n*-butyllithium and benzaldehyde in the presence of 0.77 mol % of a chiral diamine with binaphthyl skeleton gives 1-phenyl-1-pentanol in 7% *ee*.^[20] Hogeveen has also observed a rate-enhancing effect in the same reaction system, resulting in production of 1-phenyl-1-pentanol in 19% *ee* by employing 2.4 mol % of the chiral lithium amide **10** prepared from (*S*)- α -methylbenzylamine (Table 2).^[40] The rate of the catalyzed chiral reaction appears to exceed the uncatalyzed alkylation rate considerably, but the factor is too small to provide a synthetically useful turnover.

3.2. Amino Alcohol Promoted Alkylation with Diorganozinc Compounds

Among the various organometallic compounds diorganozinc acts as an ideal alkyl donor for catalytic asymmetric alkylation. In ordinary hydrocarbon or ethereal solvents, dialkylzinc compounds do not react with aldehydes. When dimethylzinc or diethylzinc, for instance, is mixed with benzaldehyde in toluene at or below room temperature, a yellow coloration occurs owing to reversible donor-acceptor complexation.^[41] However, no chemical reaction, or even change in the ¹H NMR spectrum, is observed. At elevated temperatures ethylation does take place, but very slowly.

There have been several trials using dialkylzinc for the catalytic asymmetric alkylation of benzaldehyde. In the presence of Pd^{II} or Co^{II} complexes of (1*R*)-camphorquinone oxime, the reaction is accelerated to some extent and the ethylation product is obtained with 40–60% *ee*.^[42a] In 1984 Oguni and Omi reported that certain β -amino alcohols also catalyze ethylation of benzaldehyde.^[42b] For example, the reaction occurs in toluene at 24 °C in the presence of 2 mol % of (*S*)-leucinol, giving after 48 h (*R*)-1-phenyl-1-propanol (*R*)-**18** in 48.8% *ee* and 96% yield. However, other simple amino alcohols are less efficient. In this first successful example, the *ee* value was not satisfactory and the reaction was slow.

In view of the prime significance of "ligand acceleration"^[43] in the scenario outlined in Figure 4, we have screened a variety of bidentate protic auxiliaries and aprotic ligands in the hope of obtaining greater acceleration of the nucleophilic alkylation.^[10, 13] Figure 5 illustrates the results of the reaction of benzaldehyde and diethylzinc in a

1:1.2 mole ratio performed in toluene containing 2 mol % of various additives at 0 °C for 1 h. Aprotic additives such as *N,N,N',N'*-tetramethylethylenediamine are also effective as promoters, and the efficiency appears highly dependent on the structures and substitution patterns. β -Dialkylamino alcohols act better than the corresponding *N*-monoalkyl or non-alkylated compounds. *N,N*-Dimethyleucinol is ca. three times more reactive than leucinol itself, where the acidic proton on the nitrogen may cause complications. Further impressive rate enhancement has been obtained with sterically constrained α,β -disubstituted β -dialkylamino alcohols: some cyclic compounds proved to be 10 to 100 times more effective than simple α -amino acid derived alcohols. Later, simple acyclic β -dialkylamino alcohols possessing a bulky α -substituent were found to be equally efficient.^[13, 44]

In 1986 we reported for the first time a highly enantioselective alkylation^[10] catalyzed by (–)-3-*exo*-(dimethylamino)isoborneol [(–)-DAIB **19**],^[45] a camphor-derived, sterically congested, chiral β -dialkylamino alcohol (Fig. 6).

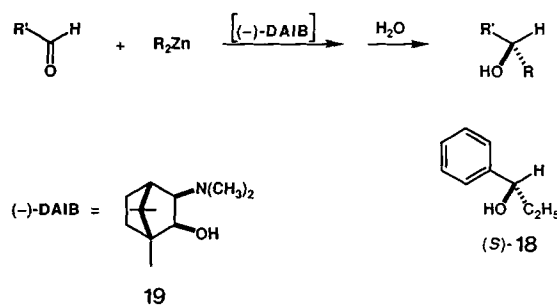


Fig. 6. DAIB-promoted enantioselective addition of dialkylzinc compounds to aldehydes. R = alkyl; R' = aryl, alkenyl, alkyl

In the presence of 2 mol % of (–)-DAIB, for example, the reaction of benzaldehyde and diethylzinc proceeds rapidly in toluene at 0 °C, and after 6 h the alcohol (*S*)-**18** is obtained in 98% *ee* and in 97% yield. With the isomer of **19**, (+)-3-*endo*-(dimethylamino)borneol **20**, the *R* alcohol in 95% *ee* is obtained.^[46]

A range of alkylating agents and aldehydes can be used for the DAIB-catalyzed enantioselective reaction. Dimethyl-, diethyl-, and other simple dialkylzinc agents are equally employable: methylation of benzaldehyde proceeds considerably more slowly than other alkylations, but gives an *ee* of greater than 90%; alcohols of type **21** are derived from substituted benzaldehydes; the reaction of furan-2-carbaldehyde and di-*n*-pentylzinc forms synthetically useful (*S*)-**22** in greater than 95% *ee*.^[13, 47] Use of ferrocenecarbaldehyde leads to optically active **23**, a key compound for the synthesis of a wide range of chiral ferrocene derivatives.^[13] Certain α,β -unsaturated or aliphatic aldehydes can also be alkylated in an enantioselective manner, in which case the *ee*'s of the products **24** are highly influenced by the substrate structures. Addition of di-*n*-pentylzinc to (*E*)-3-tributylstannylpropenal affords **25** with *S/R* = 93:7, a chiral building block of the three-component coupling prostaglandin synthesis.^[48]

Thus DAIB satisfies the conditions of an ideal chiral auxiliary that results in high reactivity and excellent stereoselectivity. Our first report^[10] was followed by publication of a

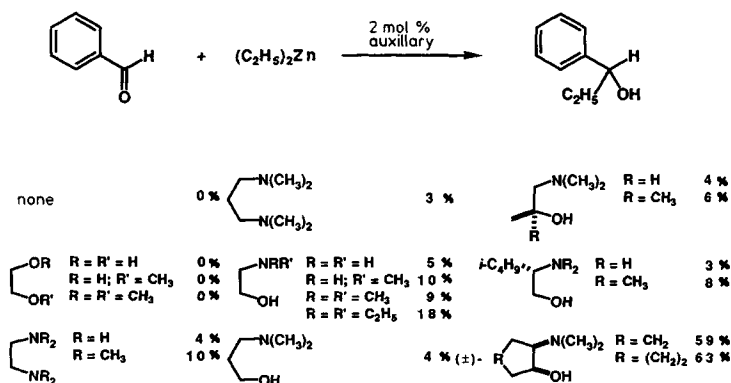
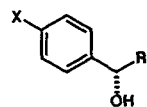
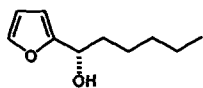


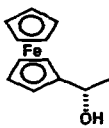
Fig. 5. Yields of ethylation of benzaldehyde with diethylzinc in the presence of 2 mol % of auxiliary in toluene at 0 °C for 1 h.



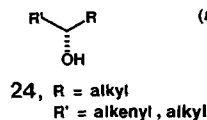
- 21a: R = CH₃, X = H
 b: R = C₂H₅, X = H
 c: R = C₂H₅, X = Cl
 d: R = C₂H₅, X = CH₃O
 e: R = *n*-C₄H₉, X = H
 f: R = C₂H₅, X = C₆H₅CO
 g: R = *n*-C₄H₉, X = C₆H₅CO



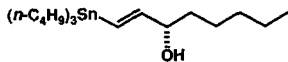
(S)-22



23



24, R = alkyl
 R' = alkenyl, alkyl



25

series of related work,^[49–52] and now a number of chiral β -amino alcohols can be used for enantioselective alkylation. Some successful examples are compiled in Table 4. The reaction of racemic 2-phenylpropanal and diethylzinc in the presence of (2*R*)-(–)-1-diisopropylamino-3,3-dimethyl-2-butanol proceeds with 5.4:1 enantiomer discrimination.^[53] We see a rather good correlation between reactivity and enantioselectivity: high enantioface selection is obtained with a fast reaction.^[13] β -Dialkylamino alcohols **26–33**, among others, are highly effective catalysts. Fréchet et al. was able to immobilize DAIB on polystyrene supports to facilitate recovery and recycling of the chiral catalyst, although this lowered the reaction rate.^[50a] Polymer-bound ephedrine acts similarly.^[50a, 51j] Some immobilized β -imino alcohols also act as catalysts.^[50b] Certain γ -dialkylamino alcohol derivatives such as **36**, as well as divinylzinc, can also be used (see Table 5).^[54]

Reactivity is increased by some functional groups near to the carbonyl group, but enantioselectivity is often diminished, probably because heteroatom coordination to the Zn atom facilitates the uncatalyzed achiral pathway. For instance, benzyloxyacetaldehyde or propyl pyruvate is readily ethylated with diethylzinc, even without amino alcohols, but the product is racemic. *n*-Butyl acetoacetate and acetophenone are not alkylated under the standard conditions. However, alkylation of γ - or δ -keto esters proceeds to give the corresponding hydroxy esters in optically active form.^[51i] *p*-Benzoylbenzaldehyde, a keto aldehyde, is ethylated chemoselectively giving the optically active secondary alcohol **21f** without alkylating the keto group.^[51m]

The choice of solvents is important. In the DAIB-catalyzed ethylation of benzaldehyde, nonpolar solvents such as toluene, hexane, ether, or their mixtures give the most satisfactory results. The reaction in THF is sluggish and the *ee* of product **18** is only 91% (98% in toluene and 99% in a 1:1 mixture of toluene and ether). The enantioselectivity of the ethylation is virtually not affected by DAIB concentration. With regard to the temperature effect, the optical yield of **18** in toluene is decreased from 98% at –20 to 0 °C to less than 95% at 50 °C.^[58] With some other amino alcohol catalysts, opposite temperature effects on the selectivity have been observed.^[51d]

Some other active catalysts include certain tridentate diamino alcohols,^[51b] amido alcohols,^[54] γ -amino alcohols,^[51d, 54] secondary amino alcohols,^[55] and lithium,^[51b, g, i] titanium,^[56] and boron^[57] salts of the appropriate amino alcohols, amines, or sulfonamides, etc. These examples are summarized in Table 5.

A major byproduct of the reaction of benzaldehyde and diethylzinc is benzyl alcohol, whose yield increases with increasing aldehyde/alkylzinc ratio. The ethylation product, ethylzinc 1-phenyl-1-propanoxide, tends to react slowly with benzaldehyde, forming propiophenone and ethylzinc benzyl-oxide. Alkyl–Zn moieties do reduce aldehydes but are not a main source of hydrides, since the reaction using dimethylzinc also forms benzyl alcohol after aqueous workup, together with acetophenone. Diisobutylzinc, however, reduces benzaldehyde, owing to the enhanced β -hydrogen reactivity.^[50a]

3.3. Mechanism of the Amino Alcohol Catalyzed Alkylation

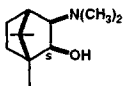
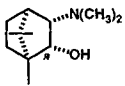
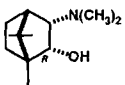
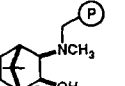
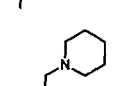
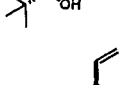
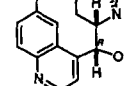
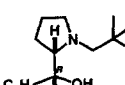
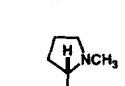
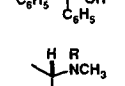
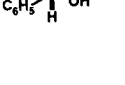
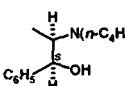
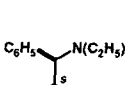
3.3.1. Transition State Models

It is worth pointing out that the stoichiometry of the aldehyde, dialkylzinc, and DAIB auxiliary has marked effects on the reactivity^[10] (Table 6). The reaction of benzaldehyde and diethylzinc does not occur in toluene at 0 °C without DAIB and, notably, no alkylation takes place with an *equimolar amount* of DAIB either. Remarkably, however, a *catalytic quantity* of DAIB promotes smooth alkylation. In other words, a complex formed from equimolar amounts of a dialkylzinc and DAIB by elimination of an alkane can not alkylate benzaldehyde despite the presence of an alkyl–Zn bond, but nevertheless does catalyze alkylation with dialkylzinc present in excess. This phenomenon clearly indicates that two Zn atoms per aldehyde are responsible for the alkyl transfer reaction. This view is also consistent with the result of a matrix isolation experiment using polymer-bound DAIB as an auxiliary.^[50a]

The exact mechanism of alkyl transfer from organometallic species to carbonyl compounds remains to be elucidated. However, various bimetallic transition state models such as **37–39** (see page 58) can explain why the highly enantioselective alkylation generates an *S* configuration.

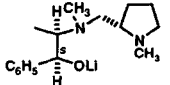
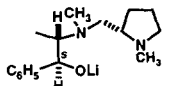
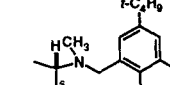
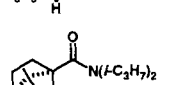
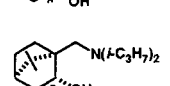
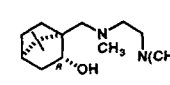
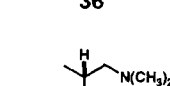
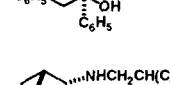
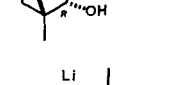
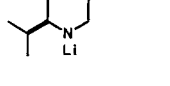
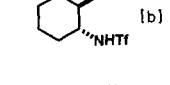
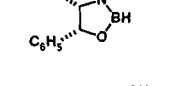
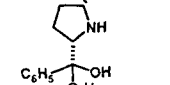
The first is the bicyclic transition state **37**.^[12] The reaction occurs from the mixed-ligand dinuclear zinc complex **40**, where DAIB structure is simplified. A number of positional isomers and stereoisomers are possible for the product-forming dinuclear complex, but the formulated structure **40** is appropriate from an electronic basis for the promotion of the alkyl transfer. In going from the ground state to the transition state, electrophilicity of the aldehyde and nucleophilicity of the alkyl group must be enhanced. As easily seen from the polar limiting structure **40b**, this electronic requirement is satisfied best by this molecular assembly. The more Lewis acidic DAIB-chelated Zn₂ accommodates the aldehyde substrate, increasing the electrophilicity at the carbon atom, and the bridging alkyl rather than the terminal alkyls

Table 4. Enantioselective addition of diorganozinc compounds to aldehydes catalyzed by β -amino alcohols [a].

Amino alcohol	Aldehyde	R ₂ Zn	Yield [%]	Alcoholic product ee [%] Configuration	Ref.
 (-)- 19	C ₆ H ₅ CHO	(CH ₃) ₂ Zn	59	91 S	[10]
	C ₆ H ₅ CHO	(C ₂ H ₅) ₂ Zn	98	99 [b] S	[10]
	C ₆ H ₅ CHO	(C ₂ H ₅) ₂ Zn	97	98 S	[10]
	C ₆ H ₅ CHO	(<i>n</i> -C ₄ H ₉) ₂ Zn	88	95 S	[13]
	<i>p</i> -ClC ₆ H ₄ CHO	(C ₂ H ₅) ₂ Zn	86	93 S	[10]
	<i>p</i> -CH ₃ OC ₆ H ₄ CHO	(C ₂ H ₅) ₂ Zn	96	93 S	[10]
	furan-2-carbaldehyde	(<i>n</i> -C ₃ H ₇) ₂ Zn	80	> 95 S	[13]
	5-methylfuran-2-carbaldehyde	(C ₂ H ₅) ₂ Zn	91	97 [c]	[47]
	ferrocenecarbaldehyde	(CH ₃) ₂ Zn	60	81 S	[13]
	(<i>E</i>)-C ₆ H ₅ CH=CHCHO	(C ₂ H ₅) ₂ Zn	81	96 S	[10]
	(<i>E</i>)-(<i>n</i> -C ₄ H ₉) ₃ SnCH=CHCHO	(<i>n</i> -C ₃ H ₇) ₂ Zn	84	85 S	[11]
 (+)- 19	C ₆ H ₅ CHO	(C ₂ H ₅) ₂ Zn	94	98 R	[13]
	C ₆ H ₅ CHO	(C ₂ H ₅) ₂ Zn	88	95 R	[46]
 20	C ₆ H ₅ CHO	(C ₂ H ₅) ₂ Zn	88	95 R	[46]
 26	C ₆ H ₅ CHO	(C ₂ H ₅) ₂ Zn	91	92 S	[50 a]
 27	C ₆ H ₅ CHO	(C ₂ H ₅) ₂ Zn	95	98 R	[13]
	(<i>E</i>)-CH ₃ CH=CHCHO	(C ₂ H ₅) ₂ Zn	90	90 R	[13]
	(<i>E</i>)-C ₆ H ₅ CH=CHCHO	(C ₂ H ₅) ₂ Zn	86	84 R	[13]
 28	<i>o</i> -C ₂ H ₅ OC ₆ H ₄ CHO	(C ₂ H ₅) ₂ Zn	72	92 [c]	[49]
 29	C ₆ H ₅ CHO	(C ₂ H ₅) ₂ Zn	100	100 R	[51 f]
	C ₆ H ₅ CH ₂ CH ₂ CHO	(C ₂ H ₅) ₂ Zn	100	86 R	[51 f]
	<i>c</i> -C ₆ H ₁₁ CHO	(C ₂ H ₅) ₂ Zn	100	86 R	[51 f]
 30	C ₆ H ₅ CHO	(C ₂ H ₅) ₂ Zn	100	99.5 [d] S	[51 f]
	C ₆ H ₅ COC ₆ H ₄ CHO	(C ₂ H ₅) ₂ Zn	99	93 S	[51 m]
	(CH ₃) ₃ SiC≡CCHO	(C ₂ H ₅) ₂ Zn	67	78 [c]	[51 k]
 6 , R = CH ₃ 31a , R = <i>i</i> -C ₃ H ₇ 31b , R = CH ₂ C ₆ H ₅ 31c , R = CH ₂ (P)	C ₆ H ₅ CHO	(C ₂ H ₅) ₂ Zn	81	81 R	[51 d]
	C ₆ H ₅ CHO	(C ₂ H ₅) ₂ Zn	72	80 R	[51 a]
	C ₆ H ₅ CHO	(C ₂ H ₅) ₂ Zn	91	86 R	[50 a]
	C ₆ H ₅ CHO	(C ₂ H ₅) ₂ Zn	96	80 R	[50 a]
 32	C ₆ H ₅ CHO	(C ₂ H ₅) ₂ Zn	100	90 S	[51 h]
	<i>n</i> -C ₆ H ₁₃ CHO	(C ₂ H ₅) ₂ Zn	95	88 S	[51 h]
	<i>n</i> -C ₆ H ₁₃ CHO	(CH ₃) ₂ Zn	70	100 S	[51 h]
	C ₆ H ₅ CH ₂ CH ₂ CHO	(C ₂ H ₅) ₂ Zn	94	95 S	[51 h]
	(CH ₃) ₂ CHCH ₂ CHO	(C ₂ H ₅) ₂ Zn	92	93 S	[51 h]
	C ₆ H ₅ COC ₆ H ₄ CHO	(C ₂ H ₅) ₂ Zn	84	91 S	[51 m]
	C ₂ H ₅ OCO(CH ₂) ₃ CHO	(C ₂ H ₅) ₂ Zn	90	92 S	[51 l]
	C ₂ H ₅ OCO(CH ₂) ₃ CHO	(C ₂ H ₅) ₂ Zn	87	95 S	[51 l]
 33	C ₆ H ₅ CHO	(C ₂ H ₅) ₂ Zn	93	94 S	[13]
 34	C ₆ H ₅ CHO	(C ₂ H ₅) ₂ Zn	97	81 S	[13]
 35	C ₆ H ₅ CHO	(C ₂ H ₅) ₂ Zn	71	49 S	[13]

[a] Most of the reactions were carried out at 0 °C in toluene or hexane. Details are found in the original papers. [b] Ether–toluene mixture as solvent. [c] Not determine [d] Lithium alkoxide of the amino alcohol was used.

Table 5. Catalytic enantioselective addition of diorganozinc reagents to aldehydes [a].

Catalyst	Aldehyde	R ₂ Zn	Yield [%]	Alcoholic product ee [%]	Configuration	Ref.
	C ₆ H ₅ CHO	(C ₂ H ₅) ₂ Zn	68	95	S	[51 b]
	C ₆ H ₅ CHO	(C ₂ H ₅) ₂ Zn	85	85	S	[51 b]
	C ₆ H ₅ CHO	(C ₂ H ₅) ₂ Zn	93	86	S	[51 c]
	C ₆ H ₅ CHO	(C ₂ H ₅) ₂ Zn	68	91	R	[54]
	C ₆ H ₅ CHO	(C ₂ H ₅) ₂ Zn	35	82	R	[54]
 36	C ₆ H ₅ CHO	(C ₂ H ₅) ₂ Zn	85	92	S	[54]
	C ₆ H ₅ CHO	(<i>n</i> -C ₃ H ₇) ₂ Zn	85	92	S	[54]
	C ₆ H ₅ CHO	(CH ₂ =CH) ₂ Zn	96	87	S	[54]
	<i>n</i> -C ₅ H ₁₁ CHO	(CH ₂ =CH) ₂ Zn	90	> 96	R	[54]
	C ₆ H ₅ CHO	(C ₂ H ₅) ₂ Zn	66	87	R	[51 d]
	C ₆ H ₅ CHO	(C ₂ H ₅) ₂ Zn	83	97	R	[55]
	C ₆ H ₅ CHO	(C ₂ H ₅) ₂ Zn	68	92	R	[51 g]
	C ₆ H ₅ CHO	(C ₂ H ₅) ₂ Zn/Ti(<i>O</i> - <i>i</i> -C ₃ H ₇) ₄	98	99	S	[56]
	C ₆ H ₅ CHO	(C ₂ H ₅) ₂ Zn	> 85	95	R	[57]
	C ₆ H ₅ CHO	(C ₂ H ₅) ₂ Zn	> 95	94	S	[51 n]
	C ₆ H ₅ CHO	(C ₂ H ₅) ₂ Zn	94	97	R	[51 o]

[a] Most of the reactions were carried out at -30°C to room temperature in toluene or hexane. Details are given in the original papers. [b] Tf = Trifluoromethanesulfonate.

acts as the migrating group. Although the electron density of the bridging R in the ground state is slightly lower than that of the terminal R, the Zn–R_{bridging} bond is more polarizable than the Zn–R_{terminal} linkage.

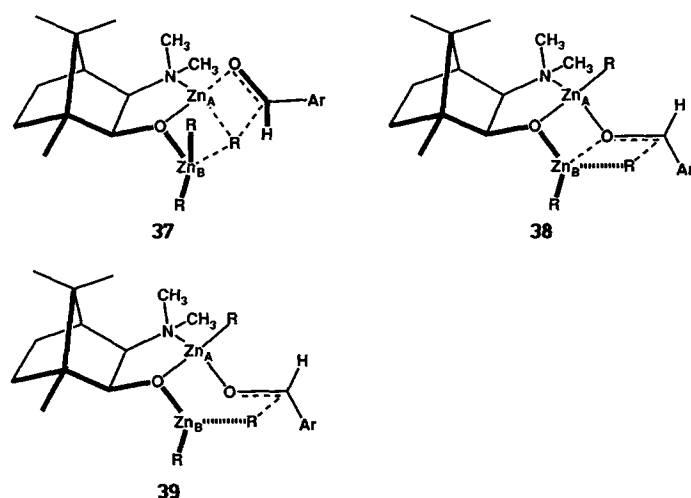
The folded bicyclic transition structure **37** features a tricoordinate structure of the migrating R group. This postulate is based on the ab initio molecular orbital calculations by *Schleyer, Houk*, and co-workers on the vapor-phase addition

Table 6. Effect of the benzaldehyde:diethylzinc:(-)-DAIB molar ratio on the yield of (*S*)-**18** [a].

Ratio $C_6H_5CHO:(C_2H_5)_2Zn:(-)-DAIB$			(<i>S</i>)-1-phenyl-1-propanol Yield [%]	<i>ee</i> [%]
1	1	0	0	—
1	1	1	1	0
1	2	2	0	—
1	2	1	88	98
2	2	1	49	96
100	50	1	48	98
50	50	1	97	98

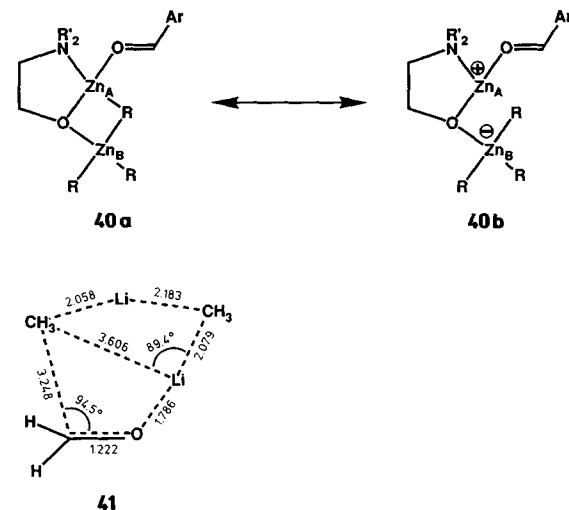
[a] All reactions were carried out under argon at 0°C in degassed anhydrous toluene.

of dimeric methyllithium to formaldehyde.^[59] The reaction is thought to involve transfer of a bridged methyl via a related planar bicyclic transition state **41**. This transition structure is considered to be very early on the reaction coordinate and the distance between the carbonyl carbon and the incoming methyl is quite large. The trajectory of attack by the



carbon nucleophile is controlled by the lithium ion, and the reacting Li–CH₃ bond is elongated to a great extent. The aggregation state of the reactive alkyllithium species in solution has been a matter of controversy,^[60] but it seems to be the consensus that dimers are the reacting entities in ethereal solvents. In THF *n*-butyllithium exists mainly as a tetramer, but the dimer is much more reactive.^[61] Monomeric species are not detected in THF.

The bicyclic transition state **38** could give an alternative explanation for the stoichiometry and stereochemistry of the organozinc reaction.^[50a] Such a mechanism involving transfer of a terminally located R to the electrophilically activated, bridged carbonyl carbon^[62] was first suggested by *Mole*



et al. for the reaction of trimethylaluminum and benzophenone.^[63] The *Pasynkiewicz–Sliwa* type bimetallic assembly **39**, postulated by *Corey* and *Hannon*,^[51c] is also in full accord with the experimental findings. This transition state model, originally proposed for the reaction of trimethylaluminum and propiophenone,^[64] is characterized by the monocyclic, six-centered structure.

3.3.2. The Sense of the Asymmetric Induction

A survey of the stereochemical outcome observed with a range of chiral β -amino alcohols as promoters (Table 4) is outlined in Figure 7.

The bicyclic transition state model of type **37** correctly predicts the absolute configuration of the products. The general sense of the asymmetric induction is controlled primarily by the chirality of the stereodetermining bicyclic intermediates, **42** and **43**.^[12] The *S*-configured Zn alkoxide is derived from **42** which possesses *S*-Zn and *S*-O bridgehead atoms, while the *R* alkoxide is obtained from the *R*-Zn, *R*-O enantiomer **43**. As illustrated by the diastereomeric transition structures **44** and **45** arising from **42**, the kinetic stereochemical bias is provided by a nonbonded repulsion between the carbonyl substituents, Ar and H, and a terminal R group attached to Zn_B. The *S*-generating geometry **44** is obviously preferable to the *R*-forming transition state **45**. The relative stabilities of the chiral bicyclic structures are influenced by substituents at the α and β positions, in accord with the general sense of Figure 7, as well as by the nitrogen substituents. The αS or βR configuration shows a preference for **42**, whereas **43** is stabilized by the enantiomeric αR or βS configuration of the auxiliary.

The degree of the enantioselection depends on the bulkiness of the substituents at the α carbon and, indeed, β -dialkyl-

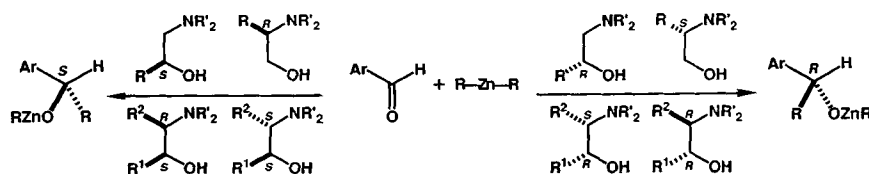
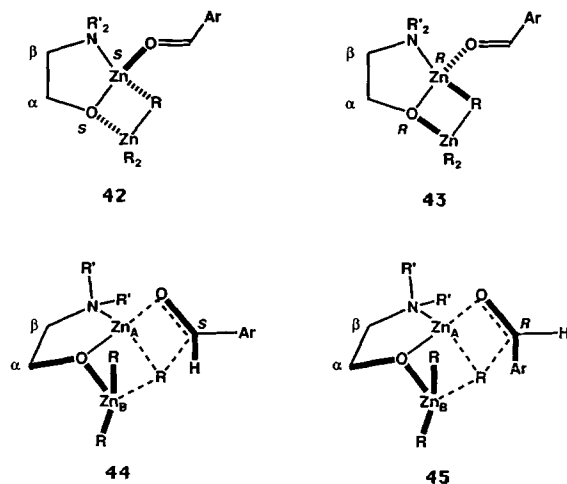


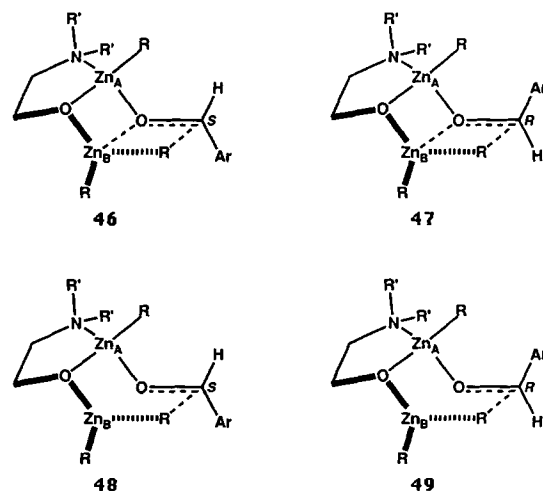
Fig. 7. General sense of asymmetric induction with a β -dialkylamino alcohol catalyst.

amino- α -*tert*-butylethanol **27** results in up to 98% *ee* in the ethylation of benzaldehyde.^[13] As is seen in (–)-DAIB, the αS , βR (or αR , βS) configuration, generating *cis*- α,β -disubstitution in the five-membered chelate complexes, is the most desirable, because the directing effects of the vicinal substituents cooperate in keeping a single chiral integrity **42** (or **43**), which presents very clear facial bias in the alkyl transfer step. With *trans*- α,β -disubstitution in the Zn chelate, the directing effects offset each other, but the α stereogenic center is more influential than the β center. In the ethylation of benzaldehyde, the diastereomeric amino alcohols, **33** and **34**, deliver the same asymmetric orientation, with higher enantioselectivity in the case of **33** (94% *ee* vs 81% *ee*, see Table 4).^[13] A similar tendency is observed with (1*R*,2*S*)-*N*-ethylephedrine and (1*R*,2*R*)-*N*-ethylpseudoephedrine (77% *ee* vs 72% *ee*).^[65] In addition, bulky alkyl groups at the nitrogen atom tend to increase the enantioselectivity.^[13]

With **35**, i.e., an (*S*)-binaphthyl moiety is the only chiral element, the ethylation of benzaldehyde proceeds in 49% optical yield (Table 4). Thus αS configured β -dialkylamino alcohols always afford *S*-enriched alkylation products, while αR auxiliaries lead predominantly to *R* products. No exceptions have been seen in the enantioselective alkylation of benzaldehyde,^[66] at least in those with a high level of stereoselection. The steric, not electronic, origin of the stereoselectivity agrees with the consistently high optical yields gained with a variety of benzaldehydes possessing electronically different substituents. Other aldehydes behave similarly. In summary, stereochemical information from the backbone of the chiral auxiliary defines the chirality of the bridgehead atoms in the stereodetermining bicyclic intermediates **42** and **43**, which in turn is transmitted to the aldehyde ligand by an alkyl transfer reaction (via **44** and **45**). Direct steric interaction between carbonyl substituents and α - or β -substituents of chiral auxiliaries is unimportant.



The same arguments on the substituent effects apply to the transition state models of the types **38**^[50a] and **39**.^[51c] Now with the αS configured amino alcohol ancillaries, the *S*-generating transition states **46** and **48** are favored over the diastereomeric *R* transition states, **47** and **49**, which suffer



significant nonbonded repulsion between Ar and the Zn_A -linked R group.^[67, 68]

3.3.3. The Reaction Pathway

On the basis of ¹H NMR studies, single-crystal X-ray analysis, molecular weight (MW) determination of certain key intermediates, alkyl scrambling experiments, kinetic measurement, etc., we have postulated the catalytic cycle of Figure 8 for the amino alcohol accelerated alkylation by dialkylzinc compounds.^[12] The scheme has a soft energy surface and compounds **50**–**53** are in dynamic equilibrium.

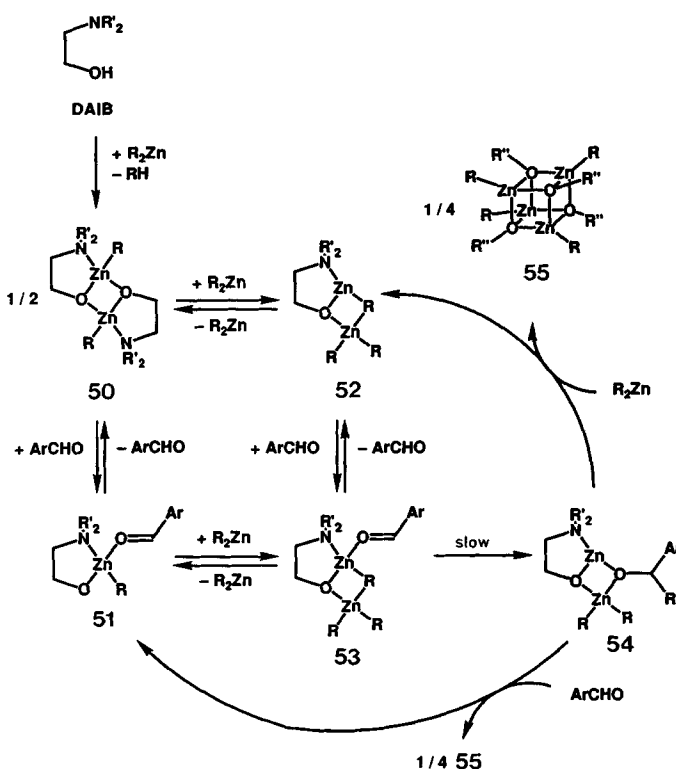


Fig. 8. Catalytic cycle of the DAIB-catalyzed enantioselective alkylation of benzaldehyde with dialkylzinc compounds R_2Zn . Ar = C_6H_5 , $R' = CH_3$, $R'' = CH(C_6H_5)R$.

In addition, alkylzinc compounds are generally flexible in nature and readily undergo interconversion with many other structural isomers by intra- or intermolecular processes: only entities with a significant contribution in the reaction are formulated in the scheme. The structural study has been done mainly by using dimethylzinc, because the methylzinc intermediates fluctuate less than the other alkyl analogues and are hence easier to monitor.

The proposed reaction pathways in Figure 8 are supported by many lines of structural information. The following phenomena are to be noted:

1) The ^1H NMR spectrum of a 1:1 mixture of benzaldehyde and dimethyl- or diethylzinc in $[\text{D}_8]\text{toluene}$ at room temperature gives only the independent signals of each component. The signals are sharp and no appreciable change in chemical shift is observed. No chemical reaction takes place under such conditions.

2) Reaction of (–)-DAIB and dimethylzinc in a 1:1 mole ratio in toluene evolves methane and produces only one dimeric compound, **50** ($\text{R} = \text{CH}_3$), among three possible stereoisomers. This complex possesses a dinuclear structure both in hydrocarbon solution and in the solid state, as substantiated by cryoscopic MW determination in benzene, ^1H and ^{13}C NMR measurements, and single-crystal X-ray analysis. The detailed structure will be given in Section 3.5 ((2*S*,2'*S*)-**50** ($\text{R} = \text{CH}_3$) in Fig. 11).

3) Complex **50** contains Zn–alkyl moieties but cannot alkylate aldehydes. However, the complex acts as a catalyst precursor. Such dimers are perhaps in equilibrium in toluene with a small amount of a coordinately unsaturated monomeric species **60**, which possesses a trigonal Zn atom, in accord with the fact that the MW decreases with decreasing concentration. This view is also supported by the results obtained with a polymer-bound DAIB **26**.^[50a]

4) The dinuclear framework of **50** ($\text{R} = \text{CH}_3$) is ruptured spontaneously upon addition of 1 equivalent of benzaldehyde, leading to the mononuclear complex **51**. The ^1H NMR spectrum at 0°C shows a broad aldehydic proton signal.^[69] This process is reversible and would occur mainly via **60** by a dissociative mechanism. The unsaturated Zn atom in the trigonal monomer **60** accepts the donor aldehyde molecule to form a tetrahedral complex **51**.^[68] MW measurement reveals that the position of the equilibrium is dependent on the **50**:aldehyde ratio.

5) The monomeric mixed ligand complex **51** is still unable to undergo alkyl transfer. The aldehyde ligand in **51** ($\text{R} = \text{C}_2\text{H}_5$; $\text{Ar} = \text{C}_6\text{H}_5$) is very slowly reduced, giving benzyl alcohol after hydrolysis. As noted in Section 3.2, only dialkylzinc compounds present in excess to DAIB or **50** can deliver alkyl groups to aldehydes.

6) Dimethylzinc also breaks the dimeric structure of **50** to give new fluctuant dimeric complexes such as **52**, exhibiting only one Zn– CH_3 signal in the ^1H NMR spectrum in a 25 to -50°C temperature range.

7) This process is also reversible as judged by MW measurement. The compound **52** ($\text{R} = \text{C}_2\text{H}_5$) and its isomers have dinuclear Zn structures and no sign of the occurrence of higher aggregates is observed in a 52 to 170 mM concentration range.

8) When 1 equivalent of benzaldehyde is introduced to the dinuclear complex **52** ($\text{R} = \text{CH}_3$) in toluene at 0°C , a new

dynamic system containing **53** is formed. The same equilibrating mixture is obtained by the reaction of the mononuclear complex **51** and dimethylzinc in a 1:1 ratio. The presence of a rapid equilibrium, $\mathbf{51} \rightleftharpoons \mathbf{53} \rightleftharpoons \mathbf{52}$, is indicated by its broad ^1H NMR spectrum giving a single broad Zn– CH_3 signal.

9) The alkyl-transfer step, $\mathbf{53} \rightarrow \mathbf{54}$, is slow. Upon standing, a 0.5:1:1 molar mixture of **50** ($\text{R} = \text{CH}_3$), dimethylzinc, and benzaldehyde at 20°C slowly yields an alkoxide assignable to **54** (MW unknown). The ^1H NMR spectrum shows a doublet at $\delta = 1.93$ and a quartet at $\delta = 5.36$ due to methyl and methine protons of the bridged alkoxide, as well as a broad singlet at $\delta = -0.17$ due to Zn– CH_3 .

10) The newly generated complex **54** ($\text{R} = \text{CH}_3$) is rather stable under these conditions and, over a period of several days, is converted slowly to the cubic Zn alkoxide tetramer **55** ($\text{R} = \text{CH}_3$) and catalyst precursor **50**. The tetrameric nature of **55** has been ascertained by cryoscopic MW measurement. Interestingly, **54** upon exposure to benzaldehyde or dimethylzinc undergoes instantaneous decomposition to **55**, completing the catalytic cycle.

Rates of the alkylation reaction are dependent on the nature of the alkyl groups. In accord with the fluctuating nature of the Zn intermediates, when two different dialkylzincs, R_1^2Zn and R_2^2Zn , are utilized in the DAIB-catalyzed reaction, a statistical distribution of the possible alkylation products, **56** and **57**, is observed (Fig. 9).^[70]

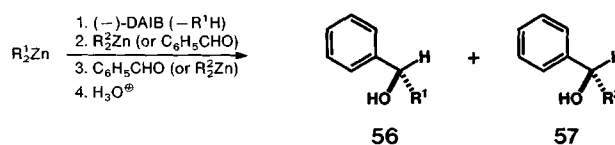


Fig. 9. Alkyl scrambling experiment.

As illustrated in Table 7, the product ratio **56**/**57** is determined only by the ratio of the two alkyl groups and their relative reactivities ($\text{CH}_3:\text{C}_2\text{H}_5:n\text{-C}_4\text{H}_9 = 1:20:8$). It is not

Table 7. Product distribution in the reaction of R_1^2Zn , R_2^2Zn , benzaldehyde, and (–)-DAIB in a 1:1:1:1 molar ratio [a].

Dialkylzincs		Product ratio		
R^1	R^2	$\text{C}_6\text{H}_5\text{CH}(\text{OH})\text{R}^1$ 56	$\text{C}_6\text{H}_5\text{CH}(\text{OH})\text{R}^2$ 57	rel. reactivity R^1/R^2
CH_3	C_2H_5	2	93	1/21
C_2H_5	CH_3	76 [b]	8 [c]	19/1
C_2H_5	$n\text{-C}_4\text{H}_9$	44	35	2.5/1
C_2H_5 [d]	$n\text{-C}_4\text{H}_9$	41	32	2.6/1
C_2H_5 [e]	$n\text{-C}_4\text{H}_9$	38	32	2.4/1
$n\text{-C}_4\text{H}_9$	C_2H_5	14	68	1/2.4
$n\text{-C}_4\text{H}_9$ [d]	C_2H_5	13	66	1/2.5
$n\text{-C}_4\text{H}_9$ [e]	C_2H_5	14	67	1/2.4

[a] R_1^2Zn was first added to a toluene solution of (–)-DAIB at 30°C , and after 15 min R_2^2Zn was added. After the mixture was stirred for 10 min and cooled to 0°C , benzaldehyde was added. The mixture was stirred at 0°C for 6 h, quenched with water, and analyzed by high pressure liquid chromatography. All reactions produced 3–14% of benzyl alcohol. [b] (S)-Product in 98% ee. [c] (S)-Product in 94% ee. [d] R_2^2Zn was added to the initially formed R_1^2Zn complex and the mixture was stirred at -78°C . Then benzaldehyde was added. [e] Benzaldehyde was added to the R_1^2Zn complex prior to addition of R_2^2Zn (see Fig. 9).

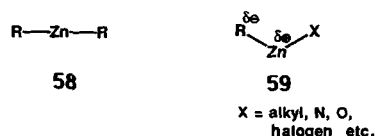
influenced by the order or conditions, such as temperature (30 to -78°C), of the mixing of benzaldehyde and the second alkylzinc to the initially formed dimeric DAIB-chelated complex **50**. The low migratory aptitude of the methyl group is consistent with the slow reaction using dimethylzinc, which may be attributed to relatively high stability of the methyl bridged dinuclear intermediates. The sense and degree of asymmetric induction in the competitive experiments are also identical with those of the independent reactions.

The mechanism of Figure 8 fully agrees with the kinetic data obtained by the DAIB-aided ethylation of benzaldehyde in toluene at 0°C . Under standard catalytic conditions, the reaction is first order in $(-)$ -DAIB but zero-order in diethylzinc and benzaldehyde. The rate is only affected by aldehyde concentrations when they are lower than 0.3 M. The kinetic results also agree with the presence of the equilibrium described in Figure 8 (complexes **50**–**53**) and, importantly, indicate that the alkyl transfer is occurring intramolecularly from the dinuclear mixed-ligand complex **53**. The result convincingly dismisses the possibility of any bimolecular mechanisms such as a reaction between the DAIB-modified organozinc species and free benzaldehyde, or nucleophilic attack of a free dialkylzinc to benzaldehyde complexed with a DAIB-chelated Zn Lewis acid. The ^1H NMR study also supports this interpretation. Since the degree of enantioselection is decreased from $S:R = 99:1$ to less than $97.5:2.5$ ^[58] by raising the reaction temperature from -20 to 50°C , the alkyl-transfer step **53**→**54** via **37** is the turnover-limiting, as well as the stereo-determining step.

3.4. Origin of the Catalyst Efficiency

The efficient catalytic enantioselective reaction is a result of the combination of the stereo-regulating ability of the well-shaped amino alcohols and the appropriate kinetics. At least three factors are suitably coupled to achieve the ideal catalytic system.

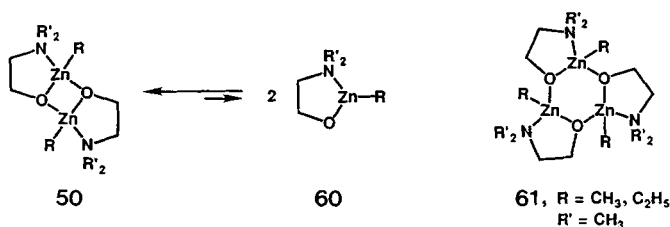
The first is the structure/reactivity profile of organozinc compounds. Monomeric dialkylzinc compounds which have an sp^2 -hybridized linear geometry at Zn (**58**) are inert to aldehydes, because the alkyl–metal linkage is rather nonpolar.^[71] However, the bond polarity can be enhanced by creating a bent geometry **59**, where the Zn atom uses orbitals having a higher p character. A coordinatively unsaturated



bent compound **59**, particularly with an electronegative substituent, has a strong donor character at the alkyl group and acceptor character at the Zn atom. Such auxiliary-induced structural perturbation increasing the reactivity toward carbonyl substrates is attained by amino alcohols such as DAIB (the argument is simplified because the actual reactive species has a dinuclear structure).

The second important factor is the tendency of the alkylzinc alkoxide product **54** to form the cubic tetrameric structure **55**, which is removed from the reaction system. The tetramer **55** neither alkylates aldehydes nor acts as a catalyst: The reaction of diethylzinc and benzaldehyde in the presence of the homologue of **55** formed from enantiomerically pure (S) -1-phenyl-1-ethanol gave, after 12 h at 0°C , 1-phenyl-1-propanol in only 2.2% yield.^[39] Fréchet's matrix isolation experiment, in which filtration of the DAIB-bound polymer before hydrolysis did not affect the yield of the alkylation product, demonstrates that the final Zn alkoxides, which are soluble in hydrocarbon, do not bind to the chiral ligands.^[50a]

Third, the use of sterically congested chiral auxiliaries is essential for obtaining high reactivity. DAIB upon reaction with organozinc compounds forms the dimeric compound **50**. By contrast, 2-dimethylaminoethanol, which possesses a similar but sterically noncongested structure, is known to give trimeric alkylzinc alkoxides of type **61**.^[72] This is the origin of the great rate enhancement for the congested compound, because the dimer dissociates more readily into the reactive monomer **60** than the trimer does. Furthermore,



DAIB-derived **52** or its isomers are dinuclear in nature, whereas tetranuclear complexes are formed by mixing diethylzinc and less congested amino alcohols, such as 2-dimethylamino-3-methyl-1-butanol, 3-dicyclohexylamino-2-propanol, and 2-dimethylamino-1-phenylethanol, in a 2:1 molar ratio.^[73]

3.5. Chiral Amplification

A number of nonclassical aspects have emerged in the amino alcohol aided enantioselective alkylation, but perhaps the most striking is the far from linear relationship between *ee* values of the chiral auxiliaries and the alcoholic products. The reaction catalyzed by DAIB^[11] or Oguni's **27**^[44] in a low *ee* leads to the alkylation products in *ee*'s which are very close to those obtained with the enantiomerically pure auxiliaries. The chirality amplifying phenomenon^[74] is clearly seen in Figure 10, which shows the *ee*'s of the *S* alcohols as a function of the *ee* of DAIB auxiliary. Typically, the reaction of benzaldehyde and diethylzinc promoted by 8 mol % of $(-)$ -DAIB in 15% *ee* in toluene at 0°C gives rise to (S) -**18** in 95% *ee* and 92% yield.

We have disclosed that the extreme convex deviation from a linear correlation results from a marked difference in chemical properties of the diastereomeric dinuclear complexes of type **50** formed from dialkylzinc and the DAIB auxiliary.^[12] When enantiomerically pure $(-)$ -DAIB is mixed with a stoichiometric quantity of dimethylzinc in

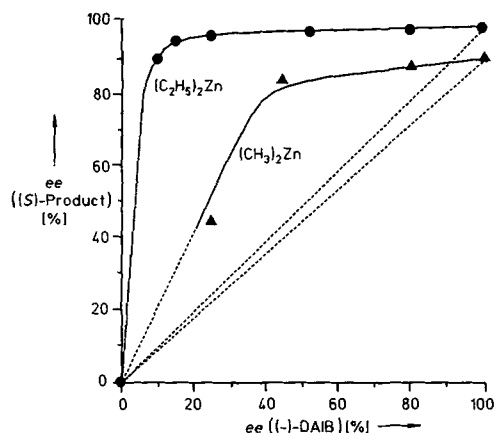


Fig. 10. The *ee* of the alkylation product as a function of the *ee* of DAIB. ● 0.42 M (C₂H₅)₂Zn, 0.42 M C₆H₅CHO, 34 mM (–)-DAIB in toluene at 0°C. ▲ 0.47 M (CH₃)₂Zn, 0.49 M C₆H₅CHO, 47 mM (–)-DAIB in [D₈]toluene at 32°C.

toluene at room temperature, a single dimeric complex, (2*S*, 2'*S*)-**50** (R = CH₃), is formed quantitatively. The stereochemistry elucidated by single crystal X-ray analysis is given in Figure 11 (top). The structure is characterized by chiral C₂ symmetry. The central Zn₂O₂ four-membered ring is "endo-fused" to the adjacent DAIB-Zn five-membered rings because of the sterically demanding bornane backbone. The central 5/4/5 tricyclic structure has *syn* geometry. The simple ¹H and ¹³C NMR spectra indicate magnetic equivalency for the two DAIB moieties of the complex in solution. The cryoscopic MW measurement also suggests a dimeric structure in benzene, but the concentration effect indicates a tendency of (2*S*,2'*S*)-**50** (R = C₂H₅) to dissociate into the monomer (2*S*)-**60**.

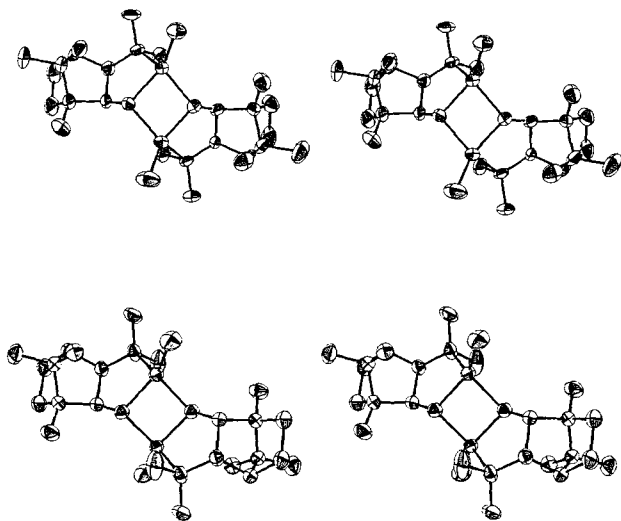


Fig. 11. Stereoview of (2*S*,2'*S*)-**50** (R = CH₃) (top) and (2*S*,2'*R*)-**50** (R = CH₃) (bottom).

Interestingly, an equimolar mixture of (2*S*,2'*S*)-**50** (R = CH₃) and the enantiomer (2*R*,2'*R*)-**50** in toluene is instantaneously converted to a new dinuclear Zn compound. The same complex is obtained by the reaction of racemic DAIB and dimethylzinc in a 1:1 mole ratio. This dinuclear

complex, (2*S*,2'*R*)-**50** (R = CH₃), is a *meso* compound and possesses achiral C_i symmetry. The ¹H and ¹³C NMR spectra are simple because of the degeneracy of the structure. The dimeric nature has been proven by MW measurement of the ethyl derivative, as the methylzinc complex has low solubility in aromatic hydrocarbons and hence MW measurement at high concentration is difficult. The single-crystal X-ray analysis (Fig. 11, bottom) of (2*S*,2'*R*)-**50** (R = CH₃) indicates that the Zn₂O₂ four-membered ring is *endo*-fused to the DAIB-Zn chelate rings, as in its chiral counterpart, but the central 5/4/5-fused tricyclic moiety has an *anti* configuration.

Thus what we have seen is strict matching of chirality or, in other words, accurate enantiomer recognition, leading to the thermodynamic situation outlined in Figure 12. Mono-

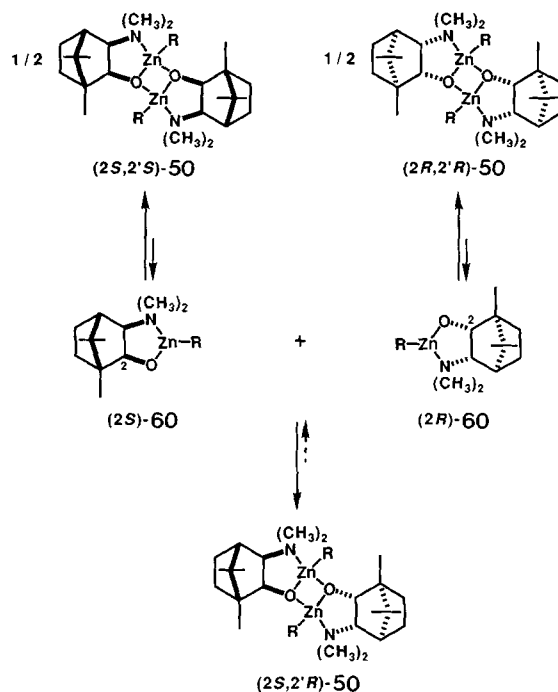


Fig. 12. Enantiomer recognition of the chiral trigonal Zn compounds **60**.

meric (2*S*)-**60** and (2*R*)-**60** are catalytically active chiral complexes derived from a dialkylzinc and (–)- or (+)-DAIB, respectively. Homochiral or heterochiral interaction between these monomers leads to the chiral dimeric complexes, (2*S*,2'*S*)-**50** and (2*R*,2'*R*)-**50**, or the *meso* dinuclear complex, (2*S*,2'*R*)-**50**, which is diastereomeric to the other two. In our system the heterochiral complex (2*S*,2'*R*)-**50** is overwhelmingly more stable than the homochiral dimers (2*S*,2'*S*)- and (2*R*,2'*R*)-**50**. When racemic DAIB is reacted with dimethylzinc in a 1:1 ratio in toluene, (2*S*,2'*R*)-**50** (R = CH₃) is produced exclusively, without forming diastereomer (2*S*,2'*S*)- and (2*R*,2'*R*)-**50**. A 1:1 mixture of (2*S*,2'*S*)- and (2*R*,2'*R*)-**50** (R = CH₃) in toluene is also instantaneously and quantitatively transformed to the more stable (2*S*,2'*R*)-**50** by way of the monomers, (2*S*)- and (2*R*)-**60**.

The chemical properties of the diastereomeric complexes, (2*S*,2'*R*)-**50** and (2*S*,2'*S*)- or (2*R*,2'*R*)-**50** are surprisingly different. The *meso* compound, (2*S*,2'*R*)-**50** (R = CH₃), is not

affected by the addition of 1 equivalent of dimethylzinc, benzaldehyde, or their mixtures. The ^1H NMR spectrum in toluene (Fig. 13b) exhibits only independent signals of these three components: the unchanged dinuclear complex and dimethylzinc afford separate $\text{Zn}-\text{CH}_3$ signals and benzaldehyde gives a sharp aldehyde signal.

The behavior of the chiral $(2S,2'S)\text{-50}$ is, however, entirely different (Fig. 13c,d). As described in Section 3.3.3, the dimeric structure is ruptured instantaneously on exposure to equimolar amounts of dimethylzinc and benzaldehyde, establishing a dynamic equilibrium among complexes **50**–**53** (Fig. 8). In the ^1H NMR spectrum (Fig. 13d), a single $\text{Zn}-\text{CH}_3$ signal and a broad aldehydic proton are observed, consistent with this view. In addition, a small amount of **54** ($\text{R} = \text{CH}_3$) is starting to emerge.

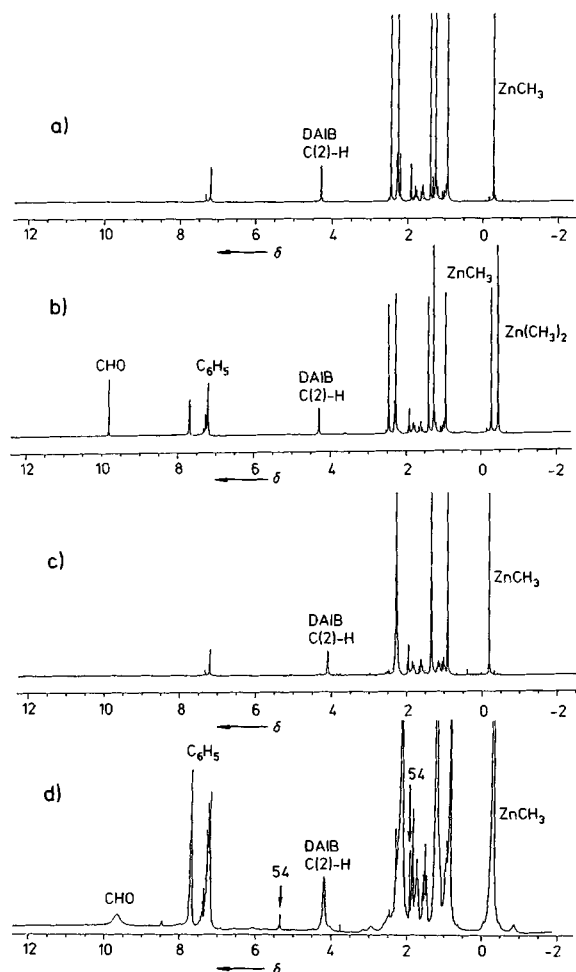


Fig. 13. ^1H NMR spectra of the dinuclear catalyst precursors and their mixtures with dimethylzinc and benzaldehyde in $[\text{D}_8]\text{toluene}$ at 25°C . a) $(2S,2'R)\text{-50}$ ($\text{R} = \text{CH}_3$). b) 0.5:1:1 $(2S,2'R)\text{-50}/(\text{CH}_3)_2\text{Zn}/\text{C}_6\text{H}_5\text{CHO}$ mixture. c) $(2S,2'S)\text{-50}$ ($\text{R} = \text{CH}_3$). d) 0.5:1:1 $(2S,2'S)\text{-50}/(\text{CH}_3)_2\text{Zn}/\text{C}_6\text{H}_5\text{CHO}$ mixture.

The ^1H NMR study using *stoichiometric* amounts of dimethylzinc, DAIB auxiliary, and benzaldehyde has thus shed light on the difference between the chiral and *meso* dinuclear complexes of type **50**: the homochiral complexes are very reactive in the presence of dialkylzincs and aldehydes, while the heterochiral isomer is unreactive. Notably,

however, under the standard *catalytic* conditions, both enantiomerically pure DAIB and the racemate do promote alkylation, but the difference in rate of these independent reactions is not large enough to explain the great chiral amplification. For example, in the ethylation using 34 mM DAIB, 0.42 M diethylzinc, and 0.42 M benzaldehyde in toluene at 0°C , the reaction with $(-)\text{-DAIB}$ is merely 14-times faster than that with $(\pm)\text{-DAIB}$. On the other hand, the reaction catalyzed by $(-)\text{-DAIB}$ in 15% *ee* under the same conditions affords the ethylation product in 95% *ee*, indicating the rate difference in the chiral and achiral pathways is about 171. The independent and competitive figures are not the same.

A refined explanation of such a phenomenon is provided by considering the difference in kinetic behavior. Figure 14 illustrates the kinetic profiles of the ethylation using $(-)\text{-DAIB}$ and racemic DAIB. The rate of the reaction using

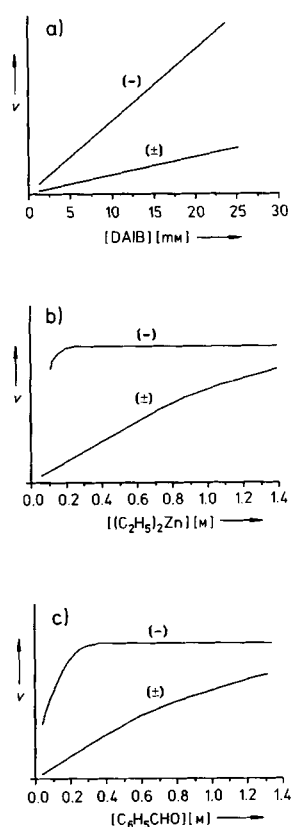


Fig. 14. Kinetic profiles of the reaction of benzaldehyde and diethylzinc catalyzed by $(-)$ - and racemic DAIB. a) The plot of the initial rate as a function of DAIB concentration (0.41 M $(\text{C}_2\text{H}_5)_2\text{Zn}$, 0.41 M $\text{C}_6\text{H}_5\text{CHO}$). b) The plot of the initial rate as a function of $(\text{C}_2\text{H}_5)_2\text{Zn}$ concentration (2.1 mM DAIB, 0.33 M $\text{C}_6\text{H}_5\text{CHO}$). c) The plot of the initial rate as a function of $\text{C}_6\text{H}_5\text{CHO}$ concentration (2.1 mM DAIB, 0.33 M $(\text{C}_2\text{H}_5)_2\text{Zn}$).

pure $(-)\text{-DAIB}$, as described above, is not affected by the concentration of diethylzinc and benzaldehyde under the ordinary conditions. By sharp contrast, the rate of the $(\pm)\text{-DAIB}$ catalyzed reaction is profoundly influenced by the dialkylzinc and aldehyde concentrations, where some bimolecular reaction between $(2S,2'R)\text{-50}$ and dialkylzinc or benzaldehyde may be involved. Therefore, the rate difference of the reaction with $(-)$ - and $(\pm)\text{-DAIB}$ becomes more

significant with lower concentrations of dialkylzincs and aldehyde substrates. In other words, with given concentrations of an organozinc and aldehyde, the larger rate difference is obtained with higher auxiliary concentration.

Such different kinetic behavior results in a marked dependence of the degree of nonlinearity on DAIB concentration. Figure 15 shows nonlinearity in the ethylation aided by (–)-DAIB in 14% *ee* [(–)/(+) = 57:43], where all reaction parameters except for DAIB concentration are fixed. Evi-

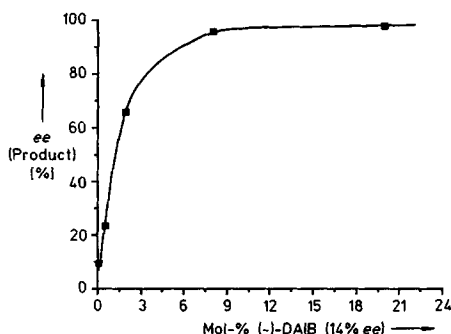


Fig. 15. Effect of auxiliary concentration on the nonlinearity in the ethylation of benzaldehyde aided by (–)-DAIB in 14% *ee* (0.42 M (C₂H₅)₂Zn, 0.42 M C₆H₅CHO, toluene, 0°C).

dently, a chiral catalyst system and achiral one are working competitively in the alkylation. When the reaction is carried out with 0.5 mol% of the partially resolved DAIB, the turnover difference is only twofold, giving the alcohol in 24% *ee*, but with the use of 20 mol% of the catalyst, the product is obtained in 98% *ee*. In the latter case, the turnover efficiency of the chiral catalyst system becomes even more than 600 times greater than that of the coexisting achiral catalyst system.^[75]

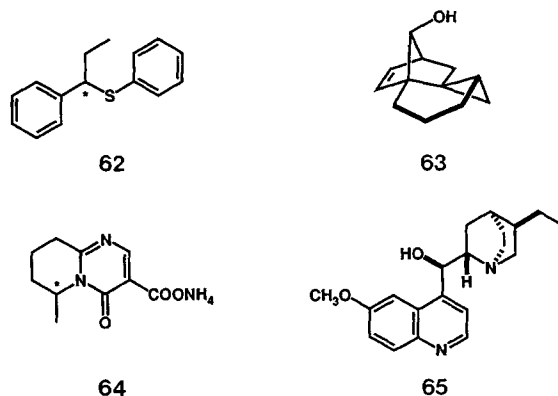
Thus we can conclude that the emergence of the intriguing nonlinearity is a result of conspicuous differences in the properties of the diastereomeric complexes, (2*S*,2'*S*)- or (2*R*,2'*R*)-**50** and (2*S*,2'*R*)-**50**. The origin of the nonlinear effect is basically the differences in the stability of these complexes, but the difference in the kinetic behavior is also important for evaluating the degree of chiral amplification. When partially resolved DAIB is employed as auxiliary, the chiral and achiral dinuclear complexes are generated in a thermodynamically controlled ratio. All the minor enantiomer, within the limit of NMR accuracy, is converted to the thermodynamically stable *meso* dinuclear complex, (2*S*,2'*R*)-**50**, while the major enantiomer, present in excess, forms the less stable chiral counterpart. The latter possesses a greater tendency to dissociate into the active monomer **60**, thereby exhibiting a much higher turnover efficiency. The difference in stability is understood by comparing the relative steric congestion of the central *syn* and *anti* 5/4/5 tricyclic ring systems. The crystalline structures of these complexes (Fig. 11) indicate that the dihedral angle of the five- and four-membered rings in the *syn* isomer, (2*S*,2'*S*)-**50** (R = CH₃), is substantially larger (ca. 20°) than in the less congested *anti* isomer, (2*S*,2'*R*)-**50** (R = CH₃).

4. Diastereomeric Interactions of Enantiomers

Diastereomeric interaction of the enantiomeric organozinc compounds has played a key role in the unusual chiral amplification. Although chirality recognition is a general phenomenon, a systematic investigation of interactions between *enantiomers* possessing an identical molecular constitution has not been made. This section deals with some examples related to this effect.

Enantiomer recognition is a fundamental principle in crystal lattice formation, giving racemates or conglomerates, and multilayer interactions.^[76] Such effects are also noted during sublimation. Kwart and Hoster found that evacuation of partially resolved **62** causes loss of optical activity owing to sublimation of the enantiomorph in excess leaving behind the less volatile racemate.^[77] Some samples represented enrichment from a starting level of 6% optical purity to 74%. Similarly optical purity of mandelic acid could be enhanced or reduced, depending on the optical purity of the starting material.^[78]

Paquette et al. found spontaneous resolution by sublimation of racemic chiral alcohol **63**.^[79] Thus evacuation of **63** to 20 Torr at 20°C for several days gave a crystal of the sublimed material, whose X-ray analysis confirmed its single mirror image form. Because the individual needles were clustered and never very large, several crystals were removed together for optical rotation measurements. Consequently there was some mixing of the two mirror image forms. Nevertheless, the sublimed crystals possessed optical activity, while the residual material did not. Interestingly, optical purity of the precipitates formed by titration of nonracemic **64** with hydrochloric acid was lower than that of original **64**.^[80]



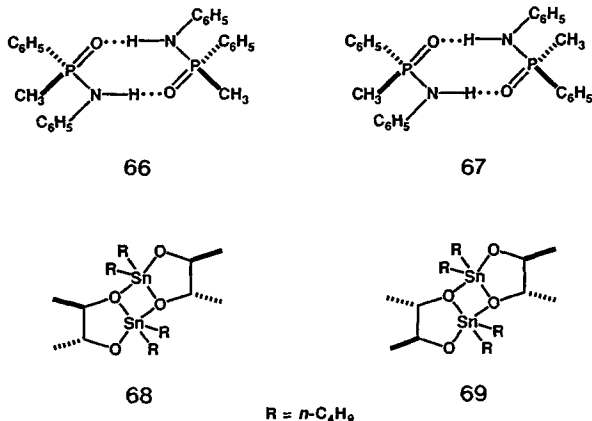
Liquid crystals are extremely sensitive to optically active compounds.^[81] Upon mixing enantiomeric cholesteric liquid crystals, a nematic phase could form.

Enantiomer recognition in solution may be interpreted by a simple symmetry argument. Under certain conditions, enantiomers can form homochiral or heterochiral associates which can be differentiated by achiral methods. Although such associates in solution are usually labile, their presence was clearly shown by Horeau et al., who found that *ee* and optical purity determined by rotation value, in general, need not be in linear relationship.^[82] The existence of such effects has also been suggested by slight differences in nonoptical

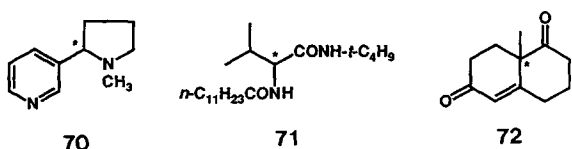
properties such as dielectric constants,^[83] boiling points,^[84] and IR spectra.^[85]

Uskoković noted that the ¹H NMR spectra of optically pure dihydroquinine **65** and of its racemate are significantly different when taken at the same concentration in CDCl₃.^[86] The spectrum of the partially resolved **65** affords two sets of peaks whose areas are proportional to the relative amount of each enantiomer. These facts can be rationalized by considering the solute-solute interactions of the enantiomers.

In a like manner, the ¹H NMR spectra of optically active, but optically impure, samples of methylphenylphosphinic anilide and related analogues exhibit distinct signals for the *P*-methyl groups in the *R* and *S* enantiomers, which may be ascribed to the molecular association through hydrogen bonding as illustrated by **66** (homochiral, favored) and **67** (heterochiral, unfavored).^[87] Similar NMR nonequivalence of the enantiomers has been noted for some substituted succinic acids and rather complex compounds having both phosphorus and carbon chiral centers.^[88] The organotin compounds derived from partially resolved 2,3-butanediol and dibutyltin oxide are shown by an ¹H NMR study to form a mixture composed of *meso* dimer **68** and chiral dimer **69**. This diastereomeric relationship may be used for the optical enrichment of the diol from 75% *ee* to 98% *ee*.^[89]

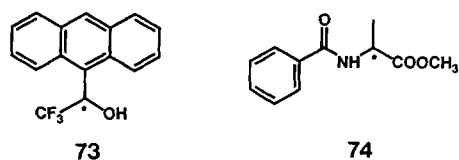


The diastereomeric self-assemblies are even separable in special environments. Thus chromatography of partially resolved enantiomers can cause depletion or enrichment in enantiomers on achiral stationary phases with an achiral mobile phase: Cundy and Crooks first resolved ¹⁴C-labeled nicotine **70** into its enantiomers with high-performance liquid chromatography on Partisil ODS and SCX through coinjection with optically active nicotine.^[90] When the diamide **71** (84% optical purity) was subjected to chromatography on Kieselgel 60 using a mixture of hexane and ethyl acetate as eluent, the elute had lower optical purity



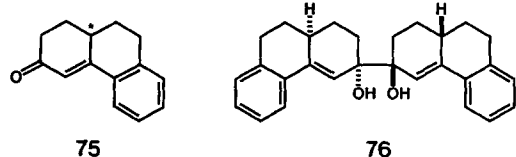
than the starting material (30% yield, 46% optical purity), and the latter fractions contained **71** with an average 90% optical purity.^[91] Under some conditions, optical enrichment from 60% to greater than 99.8% was attained. Dreiding et al. found that chromatography of a nonracemic mixture of the enantiomeric diketone **72** with achiral stationary phase (silica gel) and mobile phase (4:1 hexane-ethyl acetate) furnishes fractions which differ in *ee*.^[92]

By use of aminopropyl silica gel as the stationary phase and a hexane-2-propanol mixture as the mobile phase, chromatography of binaphthol **11** with only 33% *ee* leads to a distinct separation into two signals.^[93] The excess enantiomer is eluted first and then the racemate. The separation factor is improved on increasing the *ee* and the amount loaded. In a like manner, excess enantiomers of the chiral NMR shift reagents **73** and **74**, the tranquilizer chloromezanone, and the benzodiazepine camazepam can be separated.



The solution-phase enantiomer recognition results in various intriguing stereochemical outcomes in organic reactions:

1) In 1974, Touboul and Dana reported an amazing selectivity in the controlled potential electrochemical reduction of the enone **75**.^[94] While the dimerization of optically pure (+)-**75** or its enantiomer gave solely the *cis-threo-cis* diol (+)- and (-)-**76**, respectively, racemic **75** behaved similarly to lead to racemic **76** and to no other possible diastereomer.



2) Reductive dimerization of camphor **77** under McMurry's conditions gives the stereoisomers, **78**, **79** (and

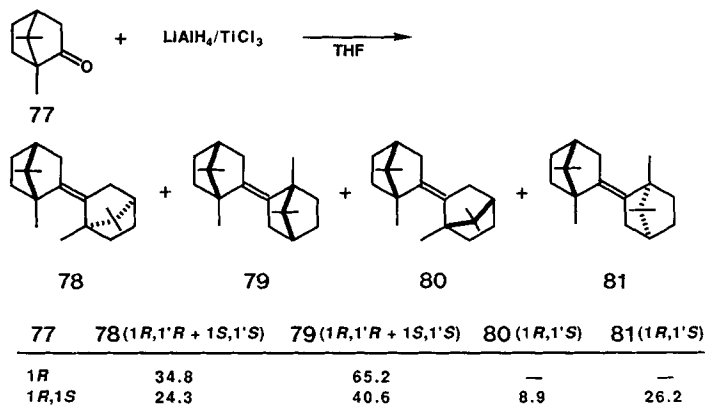


Fig. 16. Reductive dimerization of camphor **77**. Product ratios in %.

their enantiomers), **80**, and **81**. Wynberg and Feringa noticed that the reaction of racemic camphor prefers homochiral dimerization (total 64.9%) over the diastereomeric heterochiral coupling (35.1%).^[95]

3) Oxidative dimerization of the chiral phenol **82** affords the chiral dimers **83** and **84** (and their enantiomers) and the *meso* dimer **85**, whereby a significant difference in diastereoselectivity is seen between the reactions of the enantiomerically pure and racemic substrate.^[95] The reaction of (*S*)-**82** produces (*S,S*)-**83** exclusively, while the dimerization of racemic **82** lacks stereoselectivity. In the latter case, some indirect enantiomer effect assists the production of **84**, which is absent in the former reaction.

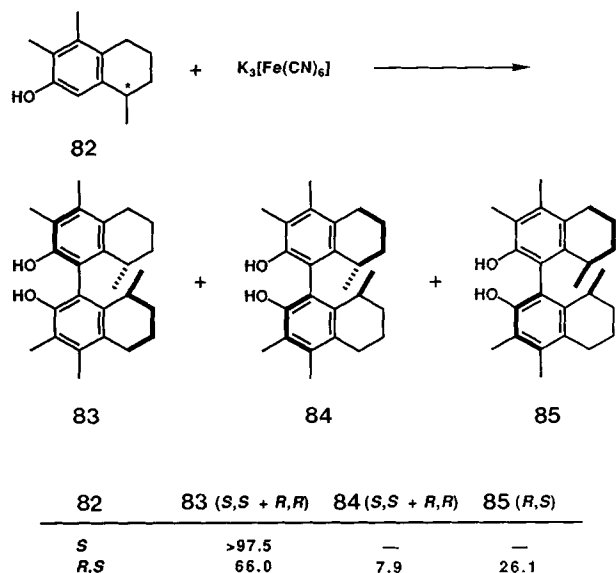


Fig. 17. Oxidative dimerization of the chiral phenol **82**. Yields in %.

4) A striking stereospecificity was observed by Gladysz et al. in the intermolecular coupling of a rhenium–carbene species.^[96] Crossover experiments using the enantiomerically configured metal complexes **86** and **87** showed that the reaction proceeds through strict enantiomer self-recognition to form only homochiral coupling products, **88** and **89** (Fig. 18).

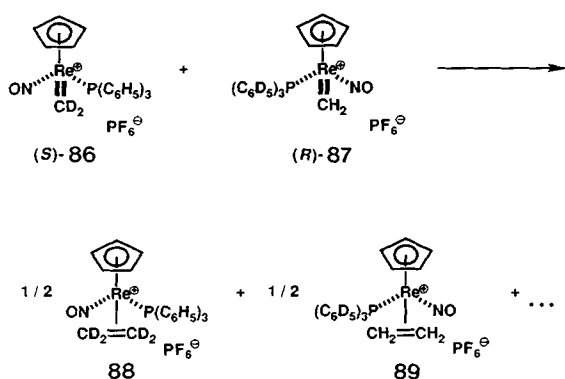


Fig. 18. Coupling of carbene ligands. The dots represent further rhenium complexes.

A considerable departure from the linear relationship between the *ee* of a chiral auxiliary and the extent of the asymmetric induction has also been observed in certain catalytic reactions:

1) Kagan, Agami, and their co-workers reported some interesting examples.^[97] First, oxidation of *p*-tolyl methyl sulfide **90** with *tert*-butyl hydroperoxide was effected in the presence of titanium tetraisopropoxide modified with enantiomerically pure (+)-diethyl tartrate (DET) to afford the (*R*)-sulfoxide **91** in 85% *ee*, whereas use of tartrate ancillary in 50% *ee* gave the product not in 42.5% *ee* as expected when linearity is assumed, but in only 19% *ee* (catalytic conditions), or 33% *ee* (stoichiometric conditions).

2) The Sharpless oxidation of geraniol **92** using enantiomerically pure DET gave the epoxide **93** in 94% *ee*. The optical yield drops to 70% *ee* by using the auxiliary with 50% *ee*. This value is considerably higher than the expected 47% *ee* (Fig. 19, center).

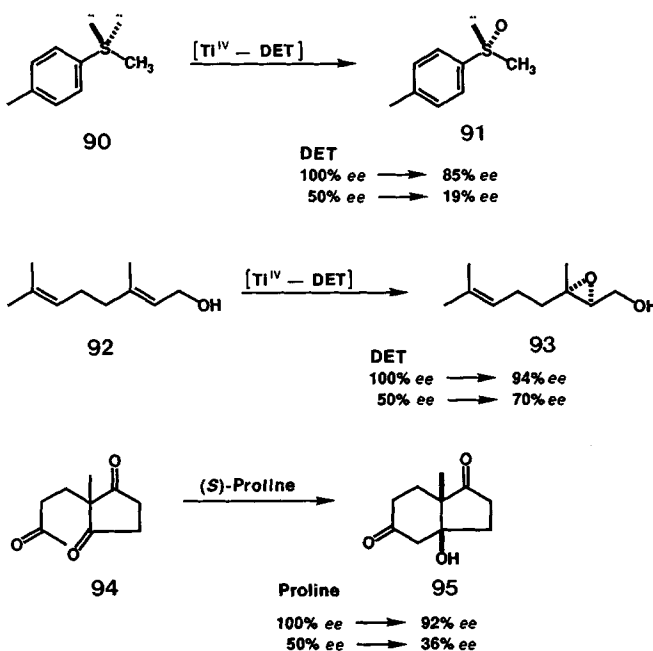


Fig. 19. Examples of the nonlinear effect in asymmetric catalysis.

3) The Hajos–Wiechert intramolecular aldol reaction of triketone **94** to **95** also exhibits a nonlinear relationship between the enantiomeric purity of the (*S*)-proline catalyst and the enantioselectivity.^[98] With the partially resolved amino acid, the cyclization gave the product in an *ee* lower than the anticipated value. On the basis of the kinetic study, these results have been interpreted in terms of the involvement of two chiral auxiliary molecules in the stereodetermining steps.^[98b]

The enantioselective addition of organozincs to prochiral aldehydes discussed in Sections 3.2 and 3.3 presents an extreme example of such a phenomenon. By virtue of the catalytic nature of the reaction together with the conspicuous difference in the chemical properties of the homochiral and heterochiral catalyst dimers, the nonlinear effect has led to unprecedentedly high chiral amplification. This molecular

mechanism may be related to the mechanism of chirality amplification in nature.^[74]

5. Outlook

Since the discovery in 1966 of asymmetric cyclopropanation of olefines with diazoalkanes, catalyzed by a chiral Schiff base/Cu^{II} complex,^[91] a variety of homogeneous asymmetric reactions using metal complex catalysts and prochiral substrates have been developed. Most of the enantioselective organometallic reactions (defined as transformations involving metal–carbon or –hydrogen bonds) rely on transition metal chemistry, but chiral multiplication can also be achieved by main group elements.^[43a, 100] This article has presented some aspects of this subject. Addition of organometallic compounds to carbonyl substrates is one of the most fundamental synthetic reactions. Although asymmetric synthesis with chirally modified organomagnesium or lithium reagents has some 50 years of history, it was only 10 years ago when a level of enantioselection greater than 90% was accomplished on the basis of stoichiometric chirality transfer. Now this domain is growing continuously, and particularly over the last five years there has been great progress in the catalytic reaction. We have realized the highly stereoselective reaction (in some cases 99% *ee*) using organozinc compounds and a catalytic quantity of chiral auxiliary, which at the same time exhibits chiral amplification. The mechanism has now been elucidated at the molecular level. This achievement, however, marks only the beginning of this important area, which is yet to be fully explored. The organozinc chemistry is intriguing, but the scope of the reaction is still limited. The procedure can be applied to a variety of organozinc compounds having primary alkyl groups, but not to the secondary and tertiary compounds, which are too unstable because of alkene elimination. Furthermore, we have not looked closely at reactions using highly functionalized organometallic reagents or substrates, which might be useful for the synthesis of complex natural products or their artificial analogues. Thus a general method for enantiofacial and selective alkylation or arylation still remains elusive. Further development of versatile methods using other metal elements is strongly desired, and we hope that the principles described here will be helpful.

In scientific research a rational approach to problems is obviously ideal, although in reality major breakthroughs in this field have been obtained mainly by serendipity or through trial-and-error improvements of the existing systems. A large body of information has accumulated in organometallic chemistry, but the level of our structural and mechanistic understanding, particularly in the supramolecular chemistry,^[9] still remains primitive. Unifying views on the structure–reactivity relationship or reliable selection rules have not appeared. However, we are quite optimistic: chemists' intuition coupled with logical designing and keen insight will continue to generate a series of versatile stereoselective organic reactions along this line.

Finally, we would emphasize that asymmetric catalysis is a four-dimensional chemistry. Stereochemical differentiation is crucially important but not enough. As illustrated

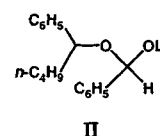
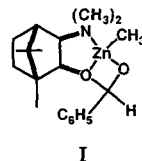
above, high efficiency is obtained only by combining the ideal three-dimensional structure (*x,y,z*) and appropriate kinetics (*t*).

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